tkutils

Tk Widgets on tclutils, 0.48.0

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1 Understanding tkutils

1.1 Introduction

tkutils provides Tk widgets and dialogs: dialogs, file list and tree, file dialog, tablelist, geometry, units, actions, launcher, designer slots and the tkdevtools app. The engines stay in tclutils, so servers and CI do not need Tk. The code names Tcl/Tk **8.6** and **9.x**.

This manual describes tkutils **0.48.0** for desktop applications built on providers and ttk. The examples and the test suite run with Tcl/Tk 8.6.14 and 9.0.4 on Linux (Xvfb). The book is not checked on Windows or macOS.

Recommended pairing: tclutils **0.68.0** + tkutils **0.48.0** + ctrlutils **0.4**. Version 0.68.0 requires nothing new from tkutils. tkutils 0.48.0 needs at least tclutils **0.64.0**: only that version brings `tusettings` (for `tkugeometry`), `tutestrun` (test runner) and the functions tkdevtools needs.

1.2 Role

Piece	Role
Prefix	package tkutils::tku<name>, namespace ::tkutils::tku<name>
Layer	view
Rule	engine in tclutils, GUI here
Loading	package require tkutils or a single module

Every module is a package `tkutils::tku<name>` in `lib/tm/tkutils/tku<name>-<version>.tm`. Its commands live in the namespace `::tkutils::tku<name>`. Most modules mark them with `namespace export`, but no module imports them into the global namespace. Therefore always call them fully qualified: `::tkutils::tkudialog::confirm`, not `tkudialog::confirm` — the short form fails with `invalid command name "tkudialog::confirm"`.

Two exceptions have a namespace of their own: tkuload lives in `::tkuload`, tkurender in `::tkurender` (chapter Designer and load). The packages are still called `tkutils::tkuload` and `tkutils::tkurender`.

1.3 The Umbrella

`package require tkutils` loads **53** modules directly, among them dialogs, file list and tree, file dialog, path bar, preview, tabs, toolbar, status bar, actions, launcher, geometry and units. Two more come in indirectly:

- tkutablelist — through `tkufilelist` (and thus `tkufiledialog`). The umbrella therefore needs **Tablelist** from tklib on the `auto_path`; without Tablelist, `package require tkutils` fails with `can't find package Tablelist`.
- `tkuopts` — the shared option check (chapter Error Messages).

After `package require tkutils`, **55** packages `tkutils::*` are loaded — 53 directly, plus `tkutablelist` 0.4 and `tkuopts` 0.1. The umbrella asks for `tkupath 0.2` and `tkupreview`

`0.1`; the versions present, 0.3 and 0.2, are loaded.

Not in the umbrella — require them one by one:

Module	Needs
tkuxml	tDOM
tkusqlite	sqlite3
tkuicon	for SVG: tksvg (Tk 8.6) or Tk 9; PNG/GIF without
tkuscrolledframe	scrollutil (tklib)
tkutical, tkumonthcanvas	tical 1.0
tkucanvaspng	tclutils::tupngdraw
tkutltools and tkutlclip, tkutlfind, tkutlfmt, tkutlfooter, tkutlsort, tkutlstate, tkutltree	Tablelist (tablelist_tile)
tkuload, tkurender	— (designer core)
tkumdview, tkupdfinspect, tkulayoutcanvas	tclutils engines

These are **20** modules; together with the 55 from the umbrella they make up the **75** modules of the tree.

```
package require tkutils
puts [package present tkutils::tkutablelist]   ;# -> 0.4, über
    tkufilelist
package require tkutils::tkuload                ;# extra
```

All 20 separate modules load when tDOM, sqlite3, tklib (Tablelist, scrollutil) and tical are on the path; `tkuicon` loads even without tksvg.

1.3.1 Adding tical

tical 1.1 is the calendar and iCalendar engine described in the book *Utils-Stack*. It is not part of tkutils, and `setup.tcl` does not find it. The tical tree ships a `pkgIndex.tcl`; add it to the `auto_path`:

```
lappend auto_path /pfad/zu/tical
package require tkutils::tkutical        ;# -> 0.3
package require tkutils::tkumonthcanvas  ;# -> 0.5
```

`tkutical` needs `tical::view::month`, `tical::view::week` and `tical::render::canvas` (1.0 each). `tkumonthcanvas` needs `tical::config`, `tical::locale`, `tical::holidays`, `tical::holidays::de` and `tical::view::month` (1.0 each).

1.3.2 One Tablelist Variant

tklib does not allow both Tablelist variants (`tablelist` and `tablelist_tile`) in one program. tkutils therefore uses **tablelist_tile** everywhere: `tkutablelist` loads it first and falls back to `Tablelist` only when it is missing. The `tkutl*` helpers load `tablelist_tile` when no

Tablelist is loaded yet; if the application already has one of the two variants, they use that one. Apps (tkdevtools, knowledge base), examples, tests and docs load `tablelist_tile`. `tkutils` followed by `tkutltools` loads without error, and so does the reverse order (tests `tkutltools-2.1`, `-2.2`). For the behavior up to 0.46.0: Appendix G.

1.4 Scope of This Book

- Not a general Tk textbook — see the German book on Tcl/Tk user interfaces
- No controllers such as `cufileops` — see *Utils-Stack*
- No engines — see *tclutils*
- Not a copy of all widget docs — the API reference is `docs/<modul>.md`

1.5 Structure

Chapter 2 covers installation and tests. Then come dialogs, file widgets, lists and the Tablelist helpers, geometry and units, actions, scaffolding, input, calendar and notes, data views, designer and load, images and tkdevtools. Chapter 16 lists the limits. The appendices hold pitfalls (A), the widget list (B), the glossary (C), error messages (D), the API catalog (E), practical examples (F) and the test environment (G).

1.6 In short

- Widgets on tclutils engines, version 0.48.0.
- Pairing tclutils 0.68.0 + tkutils 0.48.0 + ctrlutils 0.4; at least tclutils 0.64.0.
- Always call commands as `::tkutils::tku<name>::…`.
- The umbrella needs Tablelist; 20 modules come only on their own.
- `tablelist_tile` everywhere; tklib refuses both Tablelist variants in one program.

Tested with Tcl/Tk 8.6.14 (Tablelist 6.20, scrollutil 1.17) and 9.0.4 (Tablelist 7.11, scrollutil 2.10) on Linux under Xvfb, tclutils 0.68.0, tical 1.1.

Not tested: Windows, macOS.

Next chapter: Installation

2 Installation

2.1 Paths

tkutils needs tclutils on the module path, plus Tablelist from tklib (for the umbrella and all table widgets). The usual way is the bootstrap `tools/setup.tcl`:

```
source /pfad/zu/tkutils/tools/setup.tcl
package require tkutils
package require tkutils::tkudialog 0.3
```

`setup.tcl` is identical in tclutils and tkutils (`diff` shows no difference); it does not matter which copy you source — both trees are registered. The search order is as follows; only existing paths are added:

1. `TCLUTILS_TM` / `TKUTILS_TM` (override, wins)

2. `lib/tm` of its own tree
3. next to the Tcl installation (`<tcl-lib>/<paket>/lib/tm`)
4. `/usr/local/share/tcltk/<paket>/lib/tm`
5. `$XDG_DATA_HOME/tcltk/<paket>/lib/tm`, otherwise `$HOME/.local/share/tcltk/<paket>/lib/tm`
6. sibling folder `<paket>/lib/tm`, then `<paket>-*/lib/tm` (highest version)

Without `setup.tcl`, add the trees yourself:

```
tcl::tm::path add /pfad/zu/tclutils/lib/tm /pfad/zu/tkutils/lib/tm
package require tkutils::tkudialog 0.3
```

If only the tclutils tree is missing, modules that need an engine fail, for example `tkugeometry` with `can't find package tclutils::common 0.1`.

2.1.1 Copying Over an Older Tree

The umbrella of 0.48.0 is `lib/tm/tkutils-0.48.0.tm`. New module versions in 0.48.0: `tkulauncher` and `tkuload` 0.3; in 0.47.0: `tkufiledialog` 0.2; `tkutlsort`, `tkutlclip`, `tkutlfind`, `tkutltree`, `tkutltools` 0.2; `tkutlfooter`, `tkutlfmt`, `tkutlstate` 0.3. Anyone copying 0.48.0 over an older tree deletes the old files first, otherwise two versions sit side by side.

Over a 0.47.0 tree:

- `lib/tm/tkutils-0.47.0.tm`
- `lib/tm/tkutils/tkulauncher-0.2.tm`, `tkuload-0.2.tm`

Over a 0.46.0 tree, the file `lib/tm/tkutils-0.46.0.tm` instead of `tkutils-0.47.0.tm`, plus in `lib/tm/tkutils/`:

- `tkufiledialog-0.1.tm`
- `tkutlclip-0.1.tm`, `tkutlfind-0.1.tm`, `tkutlsort-0.1.tm`, `tkutltools-0.1.tm`, `tkutltree-0.1.tm`
- `tkutlfmt-0.2.tm`, `tkutlfooter-0.2.tm`, `tkutlstate-0.2.tm`
- `tkulauncher-0.2.tm`, `tkuload-0.2.tm`

The lists come from comparing the trees of 0.46.0, 0.47.0 and 0.48.0.

2.2 The Environment Variables

Tcl does **not** read TCLUTILS_TM and TKUTILS_TM itself. They take effect only where code evaluates them explicitly:

- `tools/setup.tcl` and with it the bundled apps (through `apps/_lib/paths.tcl`),
- `tkuload` on the first `build` (adds both paths with `tcl::tm::path add`),
- `tests/all.tcl`: TCLUTILS_TM helps to find the runner; the runner itself puts the trees from TKUTILS_TM and TCLUTILS_TM in front,
- the individual test files (TKUTILS_TM).

A script of your own that only calls `package require tkutils` finds nothing despite the variables being set and reports `can't find package tkutils`.

```
export TCLUTILS_TM=/pfad/zu/tclutils/lib/tm
export TKUTILS_TM=/pfad/zu/tkutils/lib/tm
```

Set them when a particular working tree should take precedence over an installation — in CI, for example.

2.3 Tests

```
xvfb-run -a tclsh tests/all.tcl
xvfb-run -a tclsh tests/all.tcl -match 'tkuunits-*'
```

The runner is tutestrun from tclutils (`::tclutils::tutestrun`, since tclutils 0.64.0). `all.tcl` looks for it in its own tree, in `TCLUTILS_TM` and in the sibling checkout `tclutils*`, never in an installation. If it is missing, `all.tcl` ends with exit code 2 and the message `all.tcl: tclutils::tutestrun not found (set TCLUTILS_TM)`.

- Every `tests/*.test` runs in a process of its own.
- The banner `Modules resolve to:` shows which files tkutils and tclutils were loaded from and names any path variables that are set.
- Limit per file: `TKUTILS_TESTGRENZE` in seconds, default 120.
- Further arguments go to tcltest (`-match`, `-skip`, ...).
- Exit code 1 when a file reports failures, writes no summary, crashes or hangs. These files are listed at the end under `Files with problems:`.

Run it with `tclsh`, not with `wish`. GUI tests carry the constraint `haveTk`; without a display they are skipped. `stack.test`, `tkumdview.test`, `tkupdfinspect.test` and `tkuwinico.test` too report `<datei>.test: no display, all tests skipped` without a display, and a summary with 0 failures.

Tests for modules whose third-party package is missing (tDOM, sqlite3, Tablelist, tical) are skipped. The tests look for tical through `TICAL_DIR` or a sibling folder. `tkutical.test` finds a folder `tical` next to `tkutils`. `tkumonthcanvas.test`, like `tkutical.test`, also finds the flat checkout; without `TICAL_DIR`, 13 of 13 tests pass there.

2.3.1 Test Run

tkutils 0.48.0 on Linux (Xvfb) with tclutils 0.68.0, tical 1.1 and `TICAL_DIR` set; Tablelist 6.20 under Tcl/Tk 8.6.14, 7.11 under 9.0.4:

Tcl/Tk	Files	Tests	passed	skipped	failed
8.6.14	78	843	831	12	0
9.0.4	78	843	840	3	0

Under 8.6.14 the suite skips the SVG tests of `tkuicon` and `tkuwinico` (no tksvg), plus one test each of `tkuscrolledframe` and `tkuwheel`. The suite is not checked on Windows or macOS. Environment and results of 0.46.0: Appendix G.

2.4 Summary

- Source `setup.tcl` or use `tcl::tm::path add` for both trees.
- TCLUTILS_TM/TKUTILS_TM work only through setup.tcl, apps, tkuload and the test run.
- Tests: `xvfb-run -a tclsh tests/all.tcl`; without a display, GUI tests are skipped.
- When copying over an older tree, delete the old `.tm` files.

Next chapter: Dialogs

3 Dialogs

3.1 tkudialog 0.3

tkudialog replaces `tk_messageBox` patterns with ttk. The message text can be selected and copied (Ctrl-C or the *Copy* button). It adds a default button, Return/Escape, focus/grab and button ids. Pure Tk, no tclutils engine.

```
package require tkutils::tkudialog 0.3
set ok [::tkutils::tkudialog::confirm "Datei löschen?" -parent .]
if {$ok} { puts "wird gelöscht" }
```

`confirm` returns **1** for *Yes*, otherwise **0** — also on Escape or when the window is closed.

3.2 All Entry Points

Call	Kind	Return value
showInfo msg ?opt?	modal	ok; "" on Escape/close
showWarning msg ?opt?	modal	like showInfo
showError msg ?opt?	modal	like showInfo; -detail starts collapsed
confirm msg ?opt?	modal	1 (Yes) or 0
input ?opt?	modal	entered text; "" on cancel or empty input
form spec ?-title? ?-parent?	modal	dict of values; "" on cancel
show win ?opt?	modal	id of the chosen button
build win ?opt?	not modal	win
choose win id	choose a button programmatically	empty
result win	state	chosen id, "" while open
getText win	state	message text
getDetail win	state	detail text
copyText win	clipboard	copied text (message + detail)

All commands live in `::tkutils::tkudialog`. There is no `message` or `warning`.

Options of `build`/`show` (and thus of the ready-made variants): `-title`, `-message`, `-icon info|warning|error|question`, `-detail`, `-buttons`, `-default`, `-cancel`, `-entry`, `-initial`, `-parent`, `-detailcollapsed`, `-width` (default 48), `-maxheight` (default 20).

`input` knows only `-title`, `-message`, `-initial`, `-parent`; `form` only `-title` and `-parent`. `input` returns `""` also for *OK* with an empty field — cancel and empty input cannot be told apart by the result.

```
package require tkutils::tkudialog 0.3
::tkutils::tkudialog::showInfo "Export fertig." -parent .
::tkutils::tkudialog::showError "Speichern fehlgeschlagen." \
    -detail "permission denied" -parent .

set name [::tkutils::tkudialog::input -title "Neuer Ordner" \
    -message "Name:" -initial "Neu" -parent .]
if {$name ne ""} { puts "anlegen: $name" }
```

3.3 Buttons with Ids

`-buttons` takes entries `Label` (id equals label, as in 0.2) or `{id Label}`. `show` returns the **id**, independent of the label — so the code stays the same when msgcat translates the labels.

```
package require tkutils::tkudialog 0.3
set id [::tkutils::tkudialog::show .frage -title "Ungespeichert" \
    -icon question -message "Änderungen speichern?" \
    -buttons {{save Speichern} {discard Verwerfen} {cancel Abbrechen}}
    \
    -default save -cancel cancel -parent .]
switch -- $id {
    save    { puts speichern }
    discard { puts verwerfen }
    default { puts abbruch }
}
```

- Return / KP_Enter chooses `-default` (without it, the first button).
- Escape and closing the window choose `-cancel`. Without it, the first button with the id `No`, `Cancel`, `no` or `cancel` or with the label `No` or `Cancel` counts; if there is none, the result is `""`.
- `confirm`, `input` and `form` use fixed ids (`yes`/`no`, `ok`/`cancel`).
- Titles and standard labels (`OK`, `Yes`, `Copy`, …) go through `msgcat` when it is loaded.
- Modal dialogs restore the caller's focus and grab.

3.4 Form

`form` embeds a `tkuform` with OK/Cancel in a modal dialog.

```
package require tkutils::tkudialog 0.3
set v [::tkutils::tkudialog::form {
    {name titel  label "Titel"    type entry}
    {name prio   label "Prio"     type combo values {niedrig normal
    hoch} default normal}
    {name fertig label "Erledigt" type check}
} -title "Neue Notiz" -parent .]
if {$v ne ""} { puts "prio=[dict get $v prio]" }
```

3.5 Not Modal

`build` builds the dialog and returns at once. You wire up the rest yourself.

```
package require tkutils::tkudialog 0.3
::tkutils::tkudialog::build .info -title "Protokoll" \
    -message "Lauf beendet." -detail "3 Dateien kopiert" \
    -buttons {{ok OK}} -default ok
puts [::tkutils::tkudialog::getDetail .info]   ;# -> 3 Dateien kopiert
::tkutils::tkudialog::copyText .info
::tkutils::tkudialog::choose .info ok
puts [::tkutils::tkudialog::result .info]      ;# -> ok
destroy .info
```

3.6 Parent and Placement

`-parent w` makes the dialog transient to the toplevel of `w` and centers it over it. Without `-parent`, the toplevel with the keyboard focus counts, otherwise`.`. Give `-parent` whenever the window is known — with several monitors it is the only reliable answer. A `-parent` that does not exist is an error `{TKUTILS TKUDIALOG PARENT <w>}` with the message `bad window path name ".nix" given as -parent` (for `-parent .nix`). A parent that is not yet visible does not make the dialog transient; it is then centered on the screen.

3.7 Options Are Checked

An unknown option is an error that lists the known ones:

```
package require tkutils::tkudialog 0.3
catch {::tkutils::tkudialog::showInfo "x" -parnet .} msg opts
puts [dict get $opts -errorcode]   ;# -> TKUTILS TKUDIALOG OPTION
    -parnet
```

The message is `unknown option "-parnet"`, with a second line `Known:` listing the valid options in the order of the code (`-title -message -icon -detail -buttons -entry -initial -parent -default -cancel -detailcollapsed -width -maxheight`). If a value is missing, it is `missing value for option "…"` with the code `{TKUTILS TKUDIALOG OPTION}`.

3.8 Versions

The file **0.2** is gone; `package require tkutils::tkudialog 0.2` loads 0.3 (compatible for plain labels). Doc: `docs/tkudialog.md`.

3.9 ctrlutils

`cufileops` has required `tkudialog` 0.3 for confirmations since 0.3; the current state in ctrlutils 0.4 is `cufileops` 0.5. See *Utils-Stack*.

3.10 Summary

- Entry points: `showInfo`, `showWarning`, `showError`, `confirm`, `input`, `form`, `build`/`show`, plus `choose`, `result`, `getText`, `getDetail`, `copyText`.
- `confirm` → 1/0, `show` → id, `input`/`form` → `""` on cancel.
- Always give `-parent`.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: all examples; the modal dialogs were closed with `choose`.

Not tested: appearance, focus and placement on a real screen.

Next chapter: File Widgets

4 File Widgets

4.1 List and Tree

Widget	Role
tkufilelist 0.2	detail list of a directory through a provider
tkufiletree 0.2	lazy tree; local or with -provider
tkupath 0.3	path bar (breadcrumb)
tkupreview 0.2	preview (text, Markdown, message, ...)

The widgets talk to a provider from `tclutils::tuprovider`, not to the fixed file system. Without `-provider`, `tkufilelist` creates a local one itself.

```
package require tclutils::tuprovider
package require tkutils::tkufilelist

set prov [::tclutils::tuprovider open local]
::tkutils::tkufilelist::widget .l -provider $prov -dir [pwd] \
    -onactivate {apply {e {
        if {[dict get $e type] eq "dir"} {
            ::tkutils::tkufilelist::setDir .l [dict get $e path]
        }
    }}}
pack .l -fill both -expand 1
::tkutils::tkufilelist::setFilter .l *.txt
```

Callbacks receive the provider's entry as a dict with `name`, `path`, `type` (`file`/`dir`), `size` and `mtime`; `size` and `mtime` may be empty. Further commands: `dir`, `refresh`, `selectedEntry`, `selectedEntries`. `tkufilelist` builds on `tkutablelist` and therefore needs Tablelist.

Options of `widget`: `-provider`, `-dir` (default `/`, not the current directory), `-files` (default 1), `-showhidden` (default 0), `-filter`, `-onactivate` (double click or Return), `-onselect`, `-height` (default 16). `setFilter` takes **one** glob pattern, case-insensitive; folders always stay visible so that navigation remains possible. Sorting by a header click keeps the mapping row → entry, because the entry is stored as a row attribute of the Tablelist.

Controllers (`cufileops`, filter, monitor) wire up behavior — book *Utils-Stack*. ZIP/remote: the provider's caps control write actions.

4.2 tkufiletree

tkufiletree reads subdirectories only when they are expanded. Options: `-root`, `-filter` (glob list, case-insensitive), `-files`, `-showhidden`, `-provider`, `-onactivate`, `-onselect`, `-height`, `-isolatekeys`.

```
package require tkutils::tkufiletree
set t [::tkutils::tkufiletree::widget .t -root [pwd] -showhidden 1 \
    -filter {*.txt *.md} \
    -onactivate {apply {p {puts "öffnen: $p"}}}]
pack $t -fill both -expand 1
::tkutils::tkufiletree::reveal $t [file join [pwd] README.md]
```

Item ids are the normalized full paths. `reveal` expands down to the target and returns 1 if exactly this target was selected, otherwise 0. Further commands: `setRoot`, `root`, `up` (also BackSpace), `refresh`, `refreshDir`, `selectedPath`, `treeview`, `volumesRoot`.

4.2.1 Hidden Files on Windows

With `-showhidden 1` the tree reads `*` and `.*`. On Windows, `*` already matches the dot files, so they came twice, and the treeview aborted with `ITEM_EXISTS`. Since 0.46.0 the matches are deduplicated with `lsort -unique` before insertion (test `tkufiletree-1.7`).

Without `-root`, the current directory counts. `volumesRoot` returns a special value for `-root`: a virtual node *This PC* with all drives. At a drive root, `up` climbs there when there is more than

one drive.

4.3 tkupath 0.3

tkupath shows a path as a row of buttons. A click sets the bar to that partial path and calls `-onnavigate` with it. The bar touches no file system; it only splits at `/`.

```
package require tkutils::tkupath
::tkutils::tkupath::widget .bar -onnavigate {apply {p {puts "gehe:
    $p"}}}
pack .bar -fill x
::tkutils::tkupath::setPath .bar /srv/daten/docs
puts [::tkutils::tkupath::getPath .bar]   ;# -> /srv/daten/docs
```

The bar has already switched before it calls the callback. The callback therefore needs no `setPath`. Only when navigation ends up elsewhere (redirect, failed change), it sets the real location with `setPath`.

Options: `-onnavigate`, `-rootlabel` (label of the root `/`, default `/`). Since 0.3 the bar recognizes drive and UNC paths; backslashes count as `/`:

Path	Segments	Targets
/srv/daten/docs	/ srv daten docs	/ /srv /srv/daten /srv/daten/docs
S:/projekte/daten	S: projekte daten	S:/ S:/projekte S:/projekte/daten
//srv/share/x	//srv/share x	//srv/share/ //srv/share/x

Up to 0.2, every target started with `/` (`/S:/projekte`); Windows reads that as a path on the current drive. `S:\projekte\daten` gives the same segments and targets as `S:/projekte/daten`. The splitting is pure string work.

4.4 tkupreview 0.2

`tkupreview` shows content that the caller supplies (`text`, `markdown`, `message` and other kinds); the app decides which file gets which kind. Since 0.2 the messages the widget writes itself (preview unavailable, file not readable, rendering failed) go through `msgcat` when it is loaded. The app puts translations into the namespace `::tkutils`:

```
package require msgcat
namespace eval ::tkutils {
    ::msgcat::mcset {} "Image preview unavailable." "Bildvorschau
    nicht verfügbar."
}
```

Without a catalog the texts stay English.

4.5 tkufiledialog 0.2

tkufiledialog is a modal dialog on `tkupath` + `tkufilelist`, in the look of the explorer stack. A local provider by default, optionally `-provider` for ZIP/remote.

Command	Return value
open ?opt?	path; with -multiple 1 a list of paths
save ?opt?	path (the file need not exist)
directory ?opt?	chosen folder, otherwise the current one

On cancel the result is `""` (or an empty list).

Option	Meaning
-parent w	toplevel for transient and placement
-title t	window title (default Open, Save, Choose folder, through msgcat)
-initialdir d	start folder; default $HOME, otherwise pwd
-provider p	tuprovider; default a fresh local one
-filetypes {{Label {.ext ...}} ...}	type combobox; filters by the first extension; without it one entry All files
-multiple 0/1	open only: several files
-showhidden 0/1	show dot files
-initialfile name	save only: preset name
-defaultextension .ext	save only: appended when the name has no extension

```
package require tkutils::tkufiledialog

set f [::tkutils::tkufiledialog::open -parent . -title "Text öffnen" \
    -initialdir [pwd] -filetypes {{Text {.txt .md}} {Alle *}}]
if {$f ne ""} { puts "gewählt: $f" }

set s [::tkutils::tkufiledialog::save -parent . -initialdir [pwd] \
    -initialfile bericht -defaultextension .txt]

set d [::tkutils::tkufiledialog::directory -parent . -initialdir [pwd]]
```

- A double click on a folder enters it.
- A double click on a file closes the dialog and returns the path (since `tkufiledialog` 0.2; in 0.1 it returned `""`).
- *OK* or Return in the name field confirms, Escape or *Cancel* cancels.
- `directory` lists folders only.
- Unknown option: `{TKUTILS TKUFILEDIALOG OPTION <opt>}`, message `unknown option "<opt>"` with a `Known:` list; unknown parent: `{TKUTILS TKUFILEDIALOG PARENT <w>}`, message `bad window path name "<w>" given as -parent`.

The double click goes through `_finishOk` like *OK* and Return (test `tkufiledialog-1.4a`). Step-by-step operation and the bug in 0.1: Appendix G.

The dialog is not a full file manager.

4.6 Summary

- List and tree talk to providers, not to a fixed file system.
- `tkufiletree -showhidden 1` shows dot files once, also on Windows.
- `tkupath` 0.3 shows drive and UNC paths with real targets.
- File dialog: `open`/`save`/`directory`, `""` on cancel; a double click on a file returns the path.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: the examples for `tkufilelist`, `tkufiletree` and `tkupath`, the table of segments and targets, `tkufiledialog.test` (7 of 7).

Not tested: Windows (hidden files in the tree, navigation with drive and UNC paths).

Next chapter: Lists and tablelist

5 Lists and tablelist

5.1 tkutablelist 0.4

A wrapper around tablelist from tklib: row/column API, sorting by header click, editable cells, selection helpers, loading and saving CSV, footer, hierarchy. Not directly in the umbrella, but it comes in through `tkufilelist`.

```
package require tkutils::tkutablelist
set w [::tkutils::tkutablelist::widget .t -columns {Name Größe Typ}]
pack $w -fill both -expand 1
::tkutils::tkutablelist::setRows $w {
    {a.txt 120 Text} {b.md 80 Markdown} {c.png 4096 Bild}
}
::tkutils::tkutablelist::selectRows $w {0 2}
puts [::tkutils::tkutablelist::selection $w]   ;# -> 0 2
```

Options of `widget`: `-columns`, `-stretch`, `-titlecolumns`, `-sortable`, `-editable`, `-selectmode` (default `extended`), `-exportselection` (default **0**), `-stripes`, `-selectcommand`, `-doublecommand`, `-editendcommand`, `-footer`. Further defaults: `-stretch all`, `-titlecolumns 0`, `-sortable 1`, `-editable 0`, `-stripes ""` (no stripes), `-footer 0`. `tableWidget` returns the raw Tablelist (path `<w>.tbl`).

5.2 Why `-exportselection 0`

Tk normally exports the selection of a list widget as the X selection PRIMARY. When another widget takes PRIMARY — a menu entry, a combobox, an entry in the toolbar —, the table loses its selection. In a file manager this means: select several files, right-click, and the selection is gone before *Copy* acts.

With exportselection 0 the selection stays. Because this changes observable behavior, the module carries the number **0.4**.

```
package require tkutils::tkutablelist
set w [::tkutils::tkutablelist::widget .alt -columns {A B} \
    -exportselection 1]
puts [[::tkutils::tkutablelist::tableWidget $w] cget -exportselection]
      ;# -> 1
```

Anyone who needs the old Tk behavior sets `-exportselection 1`.

5.3 Only 0.4

There is only **0.4**. `package require tkutils::tkutablelist 0.3` loads 0.4 (same major version 0); a separate 0.3 is no longer maintained (`package require … 0.3` returns `0.4`).

Unknown option: `{TKUTILS TKUTABLELIST OPTION -colums}` with the option name, as with `tkuopts`. The message has two lines:

```
unknown option "-colums"
Known: -columns -doublecommand -editable -editendcommand
    -exportselection -footer -selectcommand -selectmode -sortable
    -stretch -stripes -titlecolumns
```

Up to 0.46.0 the option name and the `Known:` list were missing (Appendix Pitfalls); the module version stayed 0.4. A missing value reports `{TKUTILS TKUTABLELIST OPTION}` with `missing value for option "-columns"`.

5.4 Tablelist Helpers

Sorting, display, clipboard, footer, search, column layout and tree are in the chapter Tablelist Helpers. `tkutltools` loads all seven. Wrapper and helpers load `tablelist_tile`; the order relative to the umbrella is free (chapter Understanding tkutils).

5.5 Summary

- 0.4 for multi-select + context menu; the selection survives menus.
- Old behavior with `-exportselection 1`.
- `require … 0.3` loads 0.4.
- `tkutltools` loads all seven helpers. Commands: chapter Tablelist Helpers.

Tested with Tcl/Tk 8.6.14 (Tablelist 6.20) and 9.0.4 (Tablelist 7.11) on Linux under Xvfb: all examples and messages of this chapter.

Next chapter: Tablelist Helpers

6 Tablelist Helpers

This chapter continues Lists and tablelist. The seven modules here work on a finished Tablelist, not on the wrapper `tkutablelist`. For the wrapper use `[tkutils::tkutablelist::tableWidget $w]`.

tkutltools 0.2 loads all seven and `tclutils::tunum`. It is outside the umbrella. If neither `tablelist` nor `tablelist_tile` is loaded, it requires `tablelist_tile`. If one of the two

variants is already there, it stays. Wrapper and helpers load the same variant; the order relative to the umbrella is free (chapter Understanding tkutils).

Module	Version	Task
tkutlsort	0.2	comparison per column
tkutlfmt	0.3	display per column
tkutlclip	0.2	selection as TSV or CSV
tkutlfooter	0.3	footer, sum, further aggregates
tkutlfind	0.2	search with highlighting
tkutlstate	0.3	save the column layout
tkutltree	0.2	nested dicts as a tree

`tkutltools` requires `tkutlfmt` from 0.1 and `tkutlstate` from 0.1. The files in the tree are 0.3.

6.1 Sorting: tkutlsort

```
::tkutils::tkutlsort::column  tbl spalte typ ?befehl?
::tkutils::tkutlsort::columns tbl {spalte typ spalte typ ...}
```

Type	Tablelist mode
string	dictionary
nocase	asciinocase
integer	integer
real	real
num	command through tclutils::tunum
command	own comparison with two arguments

`num` reads grouping and currency. Unreadable cells sort after the numbers. The module only sets the comparison. Sorting happens with `sortbycolumn` or a click on the column header.

An unknown type raises `{TKUTILS TKUTLSORT TYPE}`, a `command` column without a command `{TKUTILS TKUTLSORT CMD}`, an odd list for `columns` `{TKUTILS TKUTLSORT SPEC}`.

6.2 Display: tkutlfmt

```
::tkutils::tkutlfmt::column  tbl spalte typ ?option wert ...?
::tkutils::tkutlfmt::columns tbl {spalte typ ?{option wert ...}? ...}
```

The cell keeps the raw value. The display comes through `-formatcommand`. `integer`, `number`, `currency` and `percent` are right-aligned unless `-align` is given. `date` formats

with `clock`.

Option	Default	Effect
-locale	eu	eu with a dot for thousands and a comma for decimals, or us
-group	from -locale	thousands separator
-decimal	from -locale	decimal separator
-decimals	per type	decimal places: integer 0, number and currency 2, percent 1
-symbol	€	currency only
-symbolpos	post	pre or post
-align	automatic	right, left, center
-informat	empty	clock scan, otherwise a number or a recognizable time
-outformat	%Y-%m-%d	clock format

An unknown type raises `{TKUTILS TKUTLFMT TYPE}`, an unknown option `{TKUTILS TKUTLFMT OPTION -name}` and names the known ones, a foreign `-locale {TKUTILS TKUTLFMT LOCALE}` with `locale must be eu or us`.

6.3 Clipboard: tkutlclip

```
::tkutils::tkutlclip::copySelection   tbl ?option wert ...?
::tkutils::tkutlclip::copyAll         tbl ?option wert ...?
::tkutils::tkutlclip::asText          tbl zeilen ?option wert ...?
::tkutils::tkutlclip::installBindings tbl ?option wert ...?
```

`copySelection` and `copyAll` return the number of copied rows. `asText` returns the text and leaves the clipboard alone. `installBindings` binds `<<Copy>>` to the body of the table.

Option	Default	Effect
-format	tsv	tsv or csv
-header	0	column titles as the first line
-formatted	1	displayed values; with 0 the raw cells
-columns	visible	visible or all

TSV replaces tab and newline in a cell with a space. CSV keeps them. Error codes: `{TKUTILS TKUTLCLIP OPTION}` (`unknown option "-foo"`, without the option name in the code and without a `Known:` line), `FORMAT` (`format must be tsv or csv`), `COLUMNS`.

6.4 Footer: tkutlfooter

```
::tkutils::tkutlfooter::attach  haupt fuss ?-autowire 1?
::tkutils::tkutlfooter::update  haupt fuss
::tkutils::tkutlfooter::setvals fuss {wert ...}
::tkutils::tkutlfooter::autosum haupt fuss ?-columns {1 2}? ?-label
    SUM:? ?-format %.2f?
::tkutils::tkutlfooter::autoagg haupt fuss ?-columns {1 sum ...}?
    ?-label text? ?-format %.2f?
::tkutils::tkutlfooter::detach  haupt fuss
```

`attach` turns `fuss` (the footer) into a single row without selection and takes over the columns of `haupt` (the main table). `-autowire 1` chains horizontal scrolling to the existing `-xscrollcommand` of the main table instead of replacing it. Since 0.2 the footer also takes over `-titlecolumns`, hidden columns and `-stretch`.

`autosum` sums the given columns, by default all, and writes `-label` into column 0. Numbers are read by `tclutils::tunum`. After a data change, run `autosum` again. `autoagg` takes `sum`, `avg`, `min`, `max`, `count` or `countnum` per column. Counts stay integers. Errors carry `{TKUTILS TKUTLFOOTER <GRUND>}`, for example `SPEC` for an odd `-columns` list in `autoagg` and `FUNC` for an unknown aggregate. The module does not check unknown options: `attach`, `autosum` and `autoagg` skip them without an error.

6.5 Searching: tkutlfind

```
::tkutils::tkutlfind::find    tbl text ?option wert ...?
::tkutils::tkutlfind::next    tbl
::tkutils::tkutlfind::prev    tbl
::tkutils::tkutlfind::matches tbl
::tkutils::tkutlfind::clear   tbl
```

`find` returns the number of matches and colors the cells. `next` and `prev` wrap around and call `seecell`. The search runs on the displayed text, i.e. after `tkutlfmt`. After sorting, `find` is needed again, because the row numbers change. `clear` restores the background and, if set, the foreground.

Option	Default	Effect
-columns	all visible	columns
-mode	substring	substring, exact, glob, regexp
-nocase	1	ignore case
-highlightbg	#ffe9a8	background of matches
-highlightfg	empty	foreground, optional

Error codes: `{TKUTILS TKUTLFIND MODE}` (`unknown mode "xx"`) and `OPTION` (`unknown option "-foo"`, without the option name in the code).

6.6 Saving the Layout: tkutlstate

```
::tkutils::tkutlstate::save           tbl
::tkutils::tkutlstate::restore        tbl zustand
::tkutils::tkutlstate::saveToFile     tbl pfad
::tkutils::tkutlstate::restoreFromFile tbl pfad
::tkutils::tkutlstate::saveVia        tbl schlüssel setPrefix
::tkutils::tkutlstate::restoreVia     tbl schlüssel getPrefix
::tkutils::tkutlstate::autosave       tbl schlüssel setPrefix ?-delay
    ms?
::tkutils::tkutlstate::cancelAutosave  tbl
```

`save` returns a dict with `version`, `ncols`, `columns` (per column `width` and `hide`) and `sortcolumn` (`-1` without sorting). `sortorder` is added only when a column is sorted, `columnorder` only when the Tablelist knows the option `-columnorder`; Tablelist 6.20 and 7.11 do not. `restoreFromFile` and `restoreVia` return `1` when a state arrived and `0` when it is missing. `saveVia` appends key and dict to the command prefix. `restore` changes only columns that occur in both states. A value that is not a state raises `{TKUTILS TKUTLSTATE STATE}` with `not a tkutlstate dict`. An unknown option to `autosave` reports `{TKUTILS TKUTLSTATE badoption}` — the only reason in the module written in lower case.

6.7 Tree: tkutltree

A node is a dict of the fields plus an optional `children`, a list of further nodes. The top level is a list of nodes.

```
::tkutils::tkutltree::fromData tbl daten -fields {feld ...}
    ?-childrenkey children? ?-parent root?
::tkutils::tkutltree::toData   tbl       -fields {feld ...}
    ?-childrenkey children? ?-node root?
::tkutils::tkutltree::clear    tbl ?-node root?
```

Field *i* fills column *i*. A missing field becomes an empty cell. `toData` reads the raw cells, not the formatted text. A round trip gives the same structure. Error codes: `{TKUTILS TKUTLTREE FIELDS}` (`-fields is required`) and `OPTION`.

6.8 In short

- `tkutltools` 0.2 loads the seven helpers and `tunum`.
- Sorting and formatting leave the raw value in the cell.
- The footer does not replace existing horizontal scrolling.
- `tkutlstate` knows only columns that still exist.

Tested with Tcl/Tk 8.6.14 (Tablelist 6.20) and 9.0.4 (Tablelist 7.11) on Linux under Xvfb: loading `tkutltools`, the error codes and messages, `tkutlstate::save`.

Next chapter: Geometry and Units

7 Geometry and Units

7.1 tkugeometry 0.1

tkugeometry remembers size, position and zoom state of a toplevel through `tclutils::tusettings`, in a file of its own, `<Konfig-Ordner>/<app>/geometry.ini`. An app that saves its own settings dict on exit therefore cannot overwrite the geometry. Needs at least tclutils **0.64.0** (which added `tusettings`); 0.68.0 is recommended.

The configuration folder comes from `tusettings::dir config`: on Linux `$XDG_CONFIG_HOME/<app>`, otherwise `$HOME/.config/<app>`; according to the code `%APPDATA%\<app>` on Windows and `$HOME/Library/Application Support/<app>` on macOS (checked on Linux only).

Command	Effect	Return value
track w app ?-key k? ?-default geo?	restore now, save on destroy	applied geometry or ""
restore w app ?-key k? ?-default geo?	apply the saved geometry	applied geometry or ""
save w app ?-key k?	save now	saved geometry
key w	key from the path	. → main, .dlg → dlg

```
package require tkutils::tkugeometry
::tkutils::tkugeometry::track . meineapp -default 800x600         ;#
    -> 800x600
toplevel .vorschau
::tkutils::tkugeometry::track .vorschau meineapp -default 400x300 ;#
    -> 400x300
```

`track` restores by itself — a preceding `restore` is not needed. `-default` (a size such as `800x600` or a full geometry) applies when nothing is saved yet. Saving happens on the window's `<Destroy>` through a bindtag of its own; the app's `<Destroy>` bindings stay untouched. Call `save` only when the geometry should be saved at another time. With `-key` you choose a section name other than the one from `key`.

Under Xvfb without a window manager, `geometry.ini` contains after exit:

```
[main]
geometry=800x600+0+0
zoomed=0
```

7.2 Off the Screen

A saved position that is no longer on the screen — the second monitor is missing, the resolution is smaller — is not used. The window keeps its size (shrunk to the screen if necessary) and is **centered**. "On the screen" means: the top strip of the window (title bar, 100×30 pixels) lies within Tk's virtual root — on X11 all monitors as one rectangle, on Windows the virtual desktop. The check does not detect a gap between monitors of different sizes.

Example with a screen of 1280×1024 pixels: the saved value is `500x400+5000+5000`, `restore` sets `500x400+390+312` — same size, centered.

The zoom state is saved as `zoomed` and restored where Tk knows it (`wm state zoomed` on

Windows/macOS, `wm attributes -zoomed` on X11).

Errors: `{TKUTILS TKUGEOMETRY WINDOW}`, message `".nix" is not an existing toplevel` (no existing toplevel). An unknown option reports `{TCLUTILS COMMON OPTION <opt>}` with `unknown option "<opt>"` and `Known: -key -default`, because the module uses the option check of tclutils. Invalid app names and unreadable files report the `tusettings` errors, for example `{TCLUTILS TUSETTINGS APP}` with `bad app name "a/b": must be a plain directory name`.

7.3 tkuunits 0.1

tkuunits converts screen measures between px, pt, mm, cm and inch. Tk understands the units itself (`winfo fpixels. 10m`), but pixels back to millimeters or unit to unit is left to the caller. For label and print previews.

Command	Return value
units	px pt mm cm inch
toPixels wert einheit ?fenster?	pixels as a floating-point number
fromPixels px ?fenster?	dict with px pt mm cm inch
convert wert von nach ?fenster?	floating-point number

```
package require tkutils::tkuunits
puts [::tkutils::tkuunits::convert 62 mm cm]       ;# -> 6.2
puts [::tkutils::tkuunits::convert 1 inch pt]      ;# -> 72.0
set px [::tkutils::tkuunits::toPixels 62 mm]       ;# Etikettenbreite
    am Schirm
puts [dict get [::tkutils::tkuunits::fromPixels 300] mm]
```

Besides `px pt mm cm inch`, the Tk letters `p m c i` and `in` are accepted. `pt` is the printer's point, 1/72 inch, as in Tk. Pixel values depend on the resolution of the window (default`.`) and thus on `tk scaling`; unit-to-unit results do not.

Errors (messages from the test):

Code	Message
{TKUTILS TKUUNITS UNIT}	unknown unit "furlong": must be px, pt, mm, cm, inch
{TKUTILS TKUUNITS VALUE}	expected a number but got "2c"
{TKUTILS TKUUNITS WINDOW}	bad window path name ".nix"

With `tk scaling` 1.39 (Xvfb default) and 2.0, `convert 1 inch pt` stays at `72.0`; `convert 62 mm cm` returns `6.2` at 1.39 and `6.199999999999999` at 2.0 — a floating-point rounding remainder. Round output yourself, for example with `format %.2f`.

7.4 Summary

- `track` restores and saves on close; `-default` for the first start.

- Position off the screen → centered, size kept.
- Units for device-independent measures on the screen.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux, Xvfb without a window manager, screen 1280×1024: all examples, error codes and messages of this chapter.

Not tested: configuration folder on Windows and macOS, several monitors, zoom state (Xvfb without a window manager has none).

Next chapter: Actions and Layout

8 Actions and Layout

8.1 tkuaction 0.2

tkuaction describes an action once — label, command, icon, shortcut, enable/check state — and keeps all registered widgets and menu entries in sync. The module is the model; drawing happens in the widgets.

```
package require tkutils::tkuaction

proc speichern {} { puts gespeichert }
::tkutils::tkuaction::define save -label "Speichern" -command
    speichern \
    -accelerator "Ctrl+S"

foreach top {.a .b} {
    toplevel $top
    ttk::button $top.save -text Speichern \
        -command {::tkutils::tkuaction::invoke save}
    pack $top.save
    ::tkutils::tkuaction::register save $top.save
    menu $top.m -tearoff 0
    ::tkutils::tkuaction::addMenuItem $top.m save
}

::tkutils::tkuaction::setEnabled save 0 .b      ;# nur Fenster .b
puts [::tkutils::tkuaction::getEnabled save .a] ;# -> 1
puts [::tkutils::tkuaction::getEnabled save .b] ;# -> 0
::tkutils::tkuaction::setEnabled save 1         ;# global, hebt .b auf
```

8.2 Enable per Window

- `setEnabled id 0|1` without a window affects all bound widgets, as in 0.1. This call also clears all window overrides.
- `setEnabled id 0|1 $w` grays out only widgets and menu entries in the toplevel of `$w`. A child widget is resolved with `winfo toplevel`; for menus, which are toplevels themselves, the first ancestor that is not a menu counts.
- `getEnabled` and `groupSet` take the same optional window.

- `invoke` always checks the **global** flag.
- Unknown window: `{TKUTILS TKUACTION WINDOW}`.

Menu lines are not widgets; `addMenuItem $menu id` creates the line and registers its index, `registerMenu`/`unregisterMenu` bind an existing line. Groups switch several actions at once:

```
package require tkutils::tkuaction
::tkutils::tkuaction::define zoom_in  -label "Vergrößern" -command
    {puts +}
::tkutils::tkuaction::define zoom_out -label "Verkleinern" -command
    {puts -}
::tkutils::tkuaction::groupDefine bild {zoom_in zoom_out}
::tkutils::tkuaction::groupSet bild 0
```

Error codes: `{TKUTILS TKUACTION <GRUND>}` with `OPTION`, `NOACTION`, `NOGROUP`, `WINDOW`. An unknown option to `define` reports the code `{TKUTILS TKUACTION OPTION -lable}` with the option name and two lines as with `tkuopts`:

```
unknown option "-lable"
Known: -accelerator -checkable -checked -command -compound -enabled
    -icon -iconChecked -label -tooltip
```

The other messages: `unknown action 'zz'`, `unknown group 'zz'`, `unknown window '.nix'`. `getEnabled` for an unknown action returns `0` instead of an error. For the state up to 0.46.0: Appendix Pitfalls.

8.3 Scaffolding

Toolbar, status, tabs, tooltips, context menu, keys, focus, search, filter, scrolling and the selection rectangle are in the chapter Scaffolding. Forms are in the chapter Input. Viewers for CSV, JSON and diff are in the chapter Viewing Data.

8.4 tkulauncher 0.3

tkulauncher is a program launcher as a menu (`-mode menu`) or a list of buttons (`-mode list`). Entries: `app` (also `command`), `url`, `open` (also `file`), `tcl`, `system`, `menu`, `separator`, `calendar`, `calc`. URLs and files are opened by `tclutils::tuopen` with the system's default program. `app` and `tcl` entries also know `dir` (working directory) and `terminal 1` (start in a terminal window).

Options of `widget`: `-mode` (default `list`), `-text` (button text in menu mode, default `Start`), `-items`, `-file` (entries from `.json` or `.ini`), `-onlaunch`, `-settings`, `-system`, `-scroll`, `-height` (default 400), `-tooltips`, `-columns` (default 1), `-editable`.

```
package require tkutils::tkulauncher
set spec {
    {type menu label Werkzeuge items {
        {type app label Editor cmd {xterm}}
        {type url label "Tcl-Handbuch" target
    https://www.tcl.tk/man/tcl9.0/}
    }}
```

```
    {type separator}
    {type tcl label "Meine App" target app.tcl interp wish8}
}
::tkutils::tkulauncher::widget .l -mode list -items $spec \
    -onlaunch {apply {{kind argv} {puts "$kind: $argv"; return 0}}}
pack .l -fill both -expand 1
```

`-onlaunch` receives `kind argv`; a false return value suppresses the start — useful for logging, confirmation or tests. `calendar` and `calc` entries open built-in windows and do not go through `-onlaunch`.

With `-scroll 1` the button list sits in a `tkuscrolledframe` and thus needs scrollutil. The mouse wheel over the buttons scrolls the list, also over buttons that `reload` creates anew; up to 0.2 they lacked this binding. An unknown option to `widget` reports `unknown option "-foo"` with the `Known:` list from `tkuopts`.

8.4.1 Starting on Windows

On Windows, `app` entries run through `cmd /c start`, so that programs registered under *App Paths* start without a full path; the working directory goes through `start /D`. `start` reads a `/` as the beginning of a switch. Since 0.46.0 the program (first word) and the `/D` directory therefore get **backslashes**; the arguments stay as they are (they may be URLs or options). A program with `://` keeps its spelling. On Linux/macOS the program is run directly; a `dir` is set there with `cd` before the start.

8.4.2 Interpreters for `tcl` Entries

A `tcl` entry starts a script with `interp` (default `wish9`). Optionally, `tcllibpath` sets the variable `TCLLIBPATH` for the child process.

Key	Linux	Windows
wish9	wish9.0	wish90.exe
wish8	wish8.6	wish86t.exe
tclsh9	tclsh9.0	tclsh90.exe
tclsh8	tclsh8.6	tclsh86t.exe

Every platform except Windows uses the *Linux* column, macOS included.

```
package require tkutils::tkulauncher
puts [::tkutils::tkulauncher::tclInterpreters]                ;# ->
    wish9 wish8 tclsh9 tclsh8
puts [::tkutils::tkulauncher::resolveTclInterp wish8 windows] ;# ->
    wish86t.exe
::tkutils::tkulauncher::setTclInterp wish9 -linux
    /usr/local/tcl9/bin/wish9.0
puts [::tkutils::tkulauncher::resolve \
    {type tcl label Demo target app.tcl interp wish9 tcllibpath {lib}}]
```

`resolve` shows without side effects what would be started (`kind`, `argv`, optionally `env`, `dir`, `terminal`). The last line of the example prints `kind exec argv`

`{/usr/local/tcl9/bin/wish9.0 app.tcl} env {TCLLIBPATH lib}`. The launcher expands a leading `~` in paths itself, because Tcl 9 no longer does. An unknown interpreter key reports `unknown interpreter: <key>` (without an error code).

8.5 Summary

- Actions in one place; enable globally or per window (`setEnabled id 0 $w`).
- `invoke` follows the global flag.
- Launcher: Windows start with backslashes, interpreter mapping with `setTclInterp`.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: all examples and messages of this chapter, `tkulauncher.test` (100 tests).

Not tested: starting `app` entries on Windows (only read in the code), interpreters on macOS.

Next chapter: Scaffolding

9 Scaffolding

Actions and the launcher are in the chapter Actions and Layout. This chapter holds the bars, tabs, tooltips and the small helpers for focus, mouse wheel and selection rectangle. The modules belong to the umbrella, except `tkuscrolledframe`.

9.1 Toolbar: tkutoolbar

tkutoolbar 0.2 builds a bar of buttons, toggles, drop-down menus, separators and embedded widgets. Every piece has an id.

```
set tb [::tkutils::tkutoolbar::widget .tb \
    ?-orient horizontal|vertical? ?-displaymode icon|text|both? \
    ?-spacing px? ?-padding px? ?-buttonstyle flat|raised?]
::tkutils::tkutoolbar::addButton   $tb id label befehl ?-icon bild?
    ?-tooltip text? ?-shortcut taste?
::tkutils::tkutoolbar::addToggle   $tb id label varname ?-onvalue v?
    ?-offvalue v?
::tkutils::tkutoolbar::addDropdown $tb id label ?-menu {{label befehl}
    ...}?
::tkutils::tkutoolbar::addSeparator $tb
::tkutils::tkutoolbar::addWidget    $tb id kind ?seite?
```

`-buttonstyle` is `flat`. `raised` gives buttons a border. Toggles stay flat. `-shortcut` binds on the toplevel and goes away when the bar is destroyed. Tooltips go through `tkuballoon`. Without an icon, the mode `icon` shows the text.

```
::tkutils::tkutoolbar::setEnabled     $tb id 0|1
::tkutils::tkutoolbar::setDisplayMode $tb icon|text|both
::tkutils::tkutoolbar::buttonWidget   $tb id
::tkutils::tkutoolbar::items          $tb
```

Errors `{TKUTILS TKUTOOLBAR <GRUND>}`: `OPTION`, `ORIENT`, `DISPLAYMODE`, `BUTTONSTYLE`, `NOITEM`, `NOTOOLBAR`.

9.2 Status: tkustatus

tkustatus 0.2 has a main line, named fields and a progress bar. The accessors are called `setText` and `getText`, so that `set` stays free.

```
set st [::tkutils::tkustatus::widget .st]
::tkutils::tkustatus::setText   $st text
::tkutils::tkustatus::getText   $st
::tkutils::tkustatus::addField  $st id ?-width n?
::tkutils::tkustatus::setField  $st id text
::tkutils::tkustatus::fieldText $st id
::tkutils::tkustatus::progress  $st ?0..100?
::tkutils::tkustatus::flash     $st text ?ms?
```

`progress` without a value or with an empty value hides the bar. `flash` shows the text and restores the previous line afterwards.

9.3 Tabs: tkutab

tkutab 0.2 is a `ttk::notebook` with a "+" button and closable tabs. `add` returns the frame of the new tab. `current` returns the frame of the active one, `tabs` the frames in order, `count` the number.

Option	Effect
-onnew	script on "+"
-onclose	script, the frame being closed is appended
-onselect	script, the selected frame is appended

Closing works through `close` and the middle mouse button.

9.4 Tree: tkutree

tkutree 0.2 is a thin `ttk::treeview`. `tkufiletree` sits on top of it. The file window is in the chapter File Widgets.

```
set w [::tkutils::tkutree::widget .tr ?-columns ids? ?-headings texte?
    \
    ?-show spec? ?-height n? ?-command befehl?]
::tkutils::tkutree::insert   $w eltern text ?-values liste? ?-open
    0|1? ?-id id?
::tkutils::tkutree::loadTree $w knoten
::tkutils::tkutree::selection $w
::tkutils::tkutree::itemText  $w id
::tkutils::tkutree::children  $w ?id?
::tkutils::tkutree::treeview  $w
```

9.5 Tooltip: tkuballoon

tkuballoon 0.1 attaches a tooltip to any widget.

```
::tkutils::tkuballoon::add       $w text ?-delay ms? ?-wraplength px?
::tkutils::tkuballoon::clear     $w
::tkutils::tkuballoon::configure ?-delay ms? ?-wraplength px?
::tkutils::tkuballoon::cget      option
::tkutils::tkuballoon::enable
::tkutils::tkuballoon::disable
```

The defaults are 600 ms and a wrap length of 320 pixels. `disable` turns all tooltips off and hides an open one.

9.6 Context Menu: tkucontextmenu

tkucontextmenu 0.1 builds menus and binds them to the platform's context button: button 3 on X11 and Windows, button 2 and Control-click on macOS.

```
set m [::tkutils::tkucontextmenu::create .m ?-tearoff 0|1? ?-dynamic
    0|1?]
::tkutils::tkucontextmenu::addItem     $m label ?-command befehl?
    ?-accelerator text?
::tkutils::tkucontextmenu::addCheckItem $m label ?-variable v?
::tkutils::tkucontextmenu::addRadioItem $m label -variable v -value
    wert
::tkutils::tkucontextmenu::addSeparator $m
::tkutils::tkucontextmenu::addSubmenu   $m label $unter
::tkutils::tkucontextmenu::attach       $m $w ?-button 2|3?
::tkutils::tkucontextmenu::show         $m x y
```

`enable` and `disable` address an entry by its label. `createStandardEdit` creates the usual edit entries. `createFromSpec` takes a list of entries.

9.7 Keys: tkubind

tkubind 0.1 writes `Mod` as Control on X11 and Windows and as Command on macOS. Global shortcuts stay active while an entry field, a text or a Tablelist has the focus.

```
set id [::tkutils::tkubind::key "Mod-s" {speichern} ?-toplevel .? ?-id
    name?]
::tkutils::tkubind::unbind $id
::tkutils::tkubind::accelerator "Mod-Shift-p"
::tkutils::tkubind::isEditing
::tkutils::tkubind::platform ?schlüssel?
::tkutils::tkubind::context     $w {menü %X %Y}
::tkutils::tkubind::doubleClick $w {öffnen %W}
```

`accelerator` returns `Ctrl+Shift+P` on X11 and Windows, `@^P` on macOS. Groups switch several shortcuts together:

```
::tkutils::tkubind::group define editKeys {{Mod-b {fett}} {Mod-i
    {kursiv}}}
::tkutils::tkubind::group enable  editKeys ?-toplevel .?
::tkutils::tkubind::group enabled editKeys
```

`clear` removes all bindings and groups.

9.8 Focus: tkukeynav

tkukeynav 0.1 moves the focus and binds Return and Escape.

```
::tkutils::tkukeynav::next $w
::tkutils::tkukeynav::prev $w
::tkutils::tkukeynav::enable $w ?-onreturn befehl? ?-onescape befehl?
    ?-tab 0|1?
::tkutils::tkukeynav::disable $w
::tkutils::tkukeynav::form $rahmen ?-onsubmit befehl? ?-wrap 0|1?
    ?-onescape befehl?
```

`next` and `prev` return the widget that gets the focus. `form` binds the input fields in the frame.

9.9 Search and Filter

tkusearchbar 0.1 is a field with a clear button and an optional filter list. Typing calls `-command` after `-delay` milliseconds of quiet. Default 300, width 28. Clearing and a filter change fire at once. `setText` sets the text and does not fire.

```
set s [::tkutils::tkusearchbar::widget .s ?-command befehl? ?-delay
    ms? ?-filters liste?]
::tkutils::tkusearchbar::getText $s
::tkutils::tkusearchbar::clear   $s
::tkutils::tkusearchbar::getFilter $s
```

tkufilterbar 0.2 creates one field per column. `-command` receives a dict of the non-empty fields. Default pause 300 ms, field width 10. `setFilter` does not fire, `clear` does.

```
set f [::tkutils::tkufilterbar::widget .f ?-command befehl?]
::tkutils::tkufilterbar::setColumns $f {name ort}
::tkutils::tkufilterbar::getFilters $f
```

9.10 Scrolling

tkuscrolledframe 0.1 needs `scrollutil` from tklib and is outside the umbrella. `widget` returns the outer frame, `content` the frame to pack into.

```
set sf [::tkutils::tkuscrolledframe::widget .sf ?-width n? ?-height n?]
set innen [::tkutils::tkuscrolledframe::content $sf]
::tkutils::tkuscrolledframe::see $sf $kind ?ecke?
```

tkuwheel 0.2 redirects the mouse wheel from a widget to a scrollable target.

```
::tkutils::tkuwheel::redirect ziel $w ?-orient y|x|both? ?-amount n?
    ?-dynamic 0|1?
::tkutils::tkuwheel::unbind   $w ?-recursive 0|1?
```

`-dynamic 1` also covers widgets created in `$w` later.

9.11 Selection Rectangle: tkumarquee

tkumarquee 0.1 drags a rectangle on a canvas.

```
::tkutils::tkumarquee::enable  .c ?-onselect befehl? ?-onstart befehl?
    ?-button n?
::tkutils::tkumarquee::disable .c
::tkutils::tkumarquee::region  .c
::tkutils::tkumarquee::active  .c
```

`region` returns `{x1 y1 x2 y2}` or the empty string. `active` is `1` while the button is held.

9.12 In short

- `tkutoolbar` addresses widgets by id. Tooltips come from `tkuballoon`.
- `tkustatus` uses `setText`, not `set`.
- `tkutab::add` returns the content frame.
- Mod in `tkubind` is Control or Command, depending on the platform.
- `tkuscrolledframe` needs `scrollutil` and its own `package require`.

Next chapter: Input

10 Input

Forms, single labeled fields and the specialized inputs for date, time, number and tags. Validation relies on `tclutils::tuvalidate`. All modules of this chapter are in the umbrella.

10.1 Form: tkuform

tkuform 0.1 builds a form from a list of field dicts and reads the values as a dict. `tkudialog::form` uses the same module. The dialog is in the chapter Dialogs.

A field needs `name`. In addition: `label`, `type`, `default`, `values` for `combo`, `state` for `combo` (`normal` or `readonly`), `from` and `to` for `spin`, `height` for `text`. The type is `entry`, `password`, `check`, `combo`, `spin` or `text`, default `entry`.

```
set w [::tkutils::tkuform::widget .w $felder ?-padding 8? ?-columns 1?]
::tkutils::tkuform::values    $w
::tkutils::tkuform::get       $w name
::tkutils::tkuform::setField  $w name wert
::tkutils::tkuform::setValues $w dict
::tkutils::tkuform::fieldNames $w
::tkutils::tkuform::widgetOf  $w name
```

`values` returns a dict of name to value. `setValues` skips unknown keys. `setField` is named so that `set` stays free. `-columns` puts several label/field pairs side by side. A field of type `text` always takes a line of its own across the full width.

Code	Cause
{TKUTILS TKUFORM OPTION}	unknown widget option
{TKUTILS TKUFORM VALUE}	unusable -columns
{TKUTILS TKUFORM SPEC}	field without name
{TKUTILS TKUFORM TYPE}	unknown field type

10.2 One Line: tkulabeled

tkulabeled 0.1 is a line of label and control, for when a whole form is too much. The type is a value: `entry`, `combo`, `spin`, `check` or `text`.

```
::tkutils::tkulabeled::add $pfad $typ \
    ?-label text? ?-labelwidth n? ?-orient horizontal|vertical?
    ?weitere Optionen?
::tkutils::tkulabeled::value       $pfad ?neu?
::tkutils::tkulabeled::control     $pfad
::tkutils::tkulabeled::labelwidget $pfad
```

With one argument it reads the value, with two it sets it. Options other than `-label`, `-labelwidth` and `-orient` go to the inner control.

10.3 Validation: tkuvalidate

tkuvalidate 0.1 attaches a check to an entry, combo or spinbox. An invalid value turns the text red, sets the ttk state `invalid` and shows the reason through `tkuballoon`.

```
::tkutils::tkuvalidate::attach $w $prüfer \
    ?-message text? ?-when focusout|key|both? ?-allowempty 0|1? \
    ?-onvalid befehl? ?-oninvalid befehl?
::tkutils::tkuvalidate::check    $w
::tkutils::tkuvalidate::valid    $w
::tkutils::tkuvalidate::clear    $w
::tkutils::tkuvalidate::allValid {w ...}
```

`$prüfer` is a name from `tclutils::tuvalidate` (`email`, `url`, `ipv4`, `port`, `alpha`, `alnum`, `numeric`, `integer`) or a command prefix. The field's value is appended. The prefix returns a boolean. `check` and `allValid` return `0` or `1`. `clear` removes highlighting and binding.

10.4 Date: tkudateentry

tkudateentry 0.1 is a field plus a month calendar to drop down. Monday comes first. `getDate` always returns `yyyy-mm-dd`, or `""` when the field is empty or invalid. The display follows

`-dateformat`.

```
set w [::tkutils::tkudateentry::widget .d \
    ?-dateformat fmt? ?-width n? ?-textvariable var? ?-command befehl?]
::tkutils::tkudateentry::getText $w
::tkutils::tkudateentry::getDate $w
::tkutils::tkudateentry::setDate $w iso
::tkutils::tkudateentry::today   $w
::tkutils::tkudateentry::clear   $w
```

`-command` receives the ISO date, the empty string on clearing. `setDate` with a text that is not an ISO date raises `{TKUTILS TKDATEENTRY DATE}`. An empty `setDate` clears the field.

The month to click without an input line is `tkucalendar` in the chapter Calendar and Notes.

10.5 Time: tkutimeentry

tkutimeentry 0.1 shows hour and minute as spinboxes, with `-seconds` also the second. `getTime` and `setTime` use `HH:MM` or `HH:MM:SS`.

```
set w [::tkutils::tkutimeentry::widget .t \
    ?-time HH:MM? ?-increment min? ?-seconds 0|1? ?-command befehl?]
::tkutils::tkutimeentry::getTime $w
::tkutils::tkutimeentry::setTime $w 09:30
```

`-increment` is the minute step, default 5. `-command` receives the time on every change. An unusable `setTime` raises `{TKUTILS TKTIMEENTRY TIME}`.

10.6 Number: tkunumentry

tkunumentry 0.1 allows typing only a number with at most `-decimals` decimal places. Clamping and reformatting happen on commit, i.e. Return or loss of focus.

```
set w [::tkutils::tkunumentry::widget .n \
    ?-decimals n? ?-min v? ?-max v? ?-textvariable var? ?-width n?
    ?-command befehl?]
::tkutils::tkunumentry::getValue $w
::tkutils::tkunumentry::setValue $w 3.14
::tkutils::tkunumentry::clear    $w
```

`getValue` returns the number, or `""` when the field is empty. `setValue` with a text that is not a number raises `{TKUTILS TKNUMENTRY VALUE}`. `-command` receives the value on commit.

10.7 Tags: tkutags

tkutags 0.1 shows tags as removable chips. Return or "+" adds one. Duplicates and empty ones are dropped. With `-readonly`, input and remove button are missing. `-suggestions` turns the field into a combobox.

```
set w [::tkutils::tkutags::widget .t \
    ?-tags liste? ?-suggestions liste? ?-textvariable var? ?-readonly
    0|1? ?-command befehl?]
::tkutils::tkutags::getTags   $w
::tkutils::tkutags::setTags   $w liste
::tkutils::tkutags::addTag    $w tag
::tkutils::tkutags::removeTag $w tag
::tkutils::tkutags::clear     $w
```

`-textvariable` mirrors the list. If `-tags` is empty, the variable sets the initial value. `-command` receives the list on every change.

10.8 In short

- `tkuform::values` is a dict. `setField` sets one field.
- `tkuvalidate` takes the names from `tclutils::tuvalidate` or a command of your own.
- `tkudateentry::getDate` is ISO, the display follows `-dateformat`.
- `tkunumentry` formats on commit, not on every keystroke.

Next chapter: Calendar and Notes

11 Calendar and Notes

Month, week and tasks in the user interface. The engines stay in tclutils (`tucal`, `tuical`, `tunotes`, `tuvcard`, `tuldif`). `tkutical` and `tkumonthcanvas` also need tical and are outside the umbrella. How tical gets onto the `auto_path` is in the chapter Understanding tkutils; tical itself, its views and renderers are described in the book *Utils-Stack*.

11.1 A Month to Click: tkucalendar

tkucalendar 0.2 is a month of day buttons.

```
::tkutils::tkucalendar::widget       pfad ?option wert ...?
::tkutils::tkucalendar::getDate      pfad
::tkutils::tkucalendar::getFormatted pfad
::tkutils::tkucalendar::setDate      pfad iso
::tkutils::tkucalendar::today        pfad
::tkutils::tkucalendar::next         pfad
::tkutils::tkucalendar::prev         pfad
::tkutils::tkucalendar::getMonth     pfad
```

`getDate` returns the selected ISO date or `""`. `getFormatted` uses `-dateformat`. `getMonth` returns `yyyy mm` of the displayed month. `next` and `prev` page by one month.

Option	Default	Effect
-date	none	initially selected day, ISO
-firstday	monday	monday or sunday

Option	Default	Effect
-dateformat	%Y-%m-%d	format for getFormatted
-locale	the current one	names of months and weekdays
-command	empty	script, the ISO date is appended

The input line with the same calendar as a drop-down is `tkudateentry` in the chapter Input.

11.2 Text Calendar: tkucal

tkucal 0.1 shows the month as text through `tclutils::tucal`.

```
set w [::tkutils::tkucal::widget .w ?-year j? ?-month m? ?-weeknumbers
    0|1?]
::tkutils::tkucal::setMonth $w jahr monat
::tkutils::tkucal::getMonth $w
::tkutils::tkucal::next     $w
::tkutils::tkucal::prev     $w
::tkutils::tkucal::today    $w
::tkutils::tkucal::getText  $w
```

`setMonth` and `getMonth` return `{year month}`. An invalid month raises the error of `tucal`, for example `{TCLUTILS TUCAL MONTH}` with `invalid month 13`. The week starts on Sunday, as with `cal`. `getText` is the drawn calendar text.

11.3 tical: tkutical and tkumonthcanvas

tkutical 0.3 draws a month or a week through tical.

```
set w [::tkutils::tkutical::widget .w ?-year j? ?-month m? \
    ?-weeknumbers 0|1? ?-fontsize n? ?-holidays region? \
    ?-selectmode none|single|multiple? ?-command befehl?]
::tkutils::tkutical::setMonth $w jahr monat
::tkutils::tkutical::getSelection $w
::tkutils::tkutical::setSelection $w {daten und A..B}
```

`setMonth` returns `{year month}` and raises an error for an invalid month. `getSelection` returns sorted ISO dates.

tkumonthcanvas 0.5 draws month, quarter and year onto an existing canvas. `init` sets font and theme before drawing.

```
::tkutils::tkumonthcanvas::init ?-fontsize 11? ?-font Arial?
::tkutils::tkumonthcanvas::drawMonth   $c jahr monat ?x0 y0?
::tkutils::tkumonthcanvas::drawQuarter $c jahr monat ?x0 y0?
::tkutils::tkumonthcanvas::drawYear    $c jahr ?x0 y0 spalten?
::tkutils::tkumonthcanvas::clear       $c
::tkutils::tkumonthcanvas::setSelectMode none|single|multiple
::tkutils::tkumonthcanvas::getSelection
::tkutils::tkumonthcanvas::setTheme default|dark|light
::tkutils::tkumonthcanvas::addNote datum text
```

`drawQuarter` takes any month of the quarter. `setCallback` knows `day`, `week`, `month` and `select`. Themes are `default`, `dark` and `light`, plus `defineTheme`.

11.4 Appointments: tkuical

tkuical 0.2 shows the `VEVENT` entries of an iCalendar text in a table: summary, start, end, location. The engine is `tclutils::tuical`.

```
set w [::tkutils::tkuical::widget .w]
::tkutils::tkuical::loadText      $w text
::tkutils::tkuical::loadFile      $w datei
::tkutils::tkuical::setComponents $w komponenten
::tkutils::tkuical::events        $w
::tkutils::tkuical::count         $w
::tkutils::tkuical::toText        $w
::tkutils::tkuical::save          $w datei
```

`loadText` returns the number of appointments. `-editable 1` is the default and shows a bar for summary, start, end and location. `addEvent`, `removeEvent` and `setField` change an appointment. `setField` knows `summary`, `start`, `end` and `location`. Other properties and components that are not appointments are kept when writing.

11.5 Tasks: tkutodo

tkutodo 0.2 shows `VTODO` components: summary, due date, priority, percent. `-editable 1` lets the status be toggled, also with the space bar and a double click.

```
set w [::tkutils::tkutodo::widget .t ?-editable 1? ?-onchange befehl?]
::tkutils::tkutodo::loadText $w text
::tkutils::tkutodo::setTodos $w komponenten
::tkutils::tkutodo::todos    $w
::tkutils::tkutodo::count    $w
::tkutils::tkutodo::toggle   $w ?index?
```

Done sets `STATUS` to `COMPLETED`, `PERCENT-COMPLETE` to `100` and a timestamp `COMPLETED`. Back sets `NEEDS-ACTION`, `PERCENT-COMPLETE` to `0` and removes `COMPLETED`.

11.6 Notes: tkunotes

tkunotes 0.2 is a tree on the left and an editor on the right. Storage and search come from `tclutils::tunotes`.

```
set w [::tkutils::tkunotes::widget .w ?-file notizen.json? ?-toolbar
    0?]
::tkutils::tkunotes::addRoot  $w titel inhalt ?tags?
::tkutils::tkunotes::addChild $w eltern titel inhalt ?tags?
::tkutils::tkunotes::commit   $w ?id?
::tkutils::tkunotes::delete   $w id ?cascade?
::tkutils::tkunotes::search   $w text
::tkutils::tkunotes::load     $w datei
::tkutils::tkunotes::save     $w ?datei?
::tkutils::tkunotes::store    $w
```

`addRoot` returns the id. `search` returns ids. `load` returns the count. `commit` writes the editor fields into the store. `store` is the dict of `tunotes`. `-toolbar 0` is the default.

11.7 Contacts and LDIF

tkuvcard 0.2 shows every card as a node with `FN` and the properties as children. The engine is `tclutils::tuvcard`.

```
set w [::tkutils::tkuvcard::widget .w]
::tkutils::tkuvcard::loadText $w text
::tkutils::tkuvcard::cards    $w
::tkutils::tkuvcard::count    $w
```

`loadText` returns the number of cards. `-editable 1` is the default.

tkuldif 0.2 shows every LDIF entry as a node with `dn` and every attribute line as a child. The engine is `tclutils::tuldif`. `loadText` returns the number of entries. `-editable 1` is the default. `entries` returns the entries.

11.8 Calculator: tkucalc

tkucalc 0.2 is a pocket calculator. `evaluate` computes without a window; the package requires Tk anyway.

```
::tkutils::tkucalc::widget       pfad ?-onresult befehl? ?-history 0?
::tkutils::tkucalc::evaluate     text
::tkutils::tkucalc::clear        pfad
::tkutils::tkucalc::getHistory   pfad
::tkutils::tkucalc::clearHistory pfad
```

`evaluate` replaces the German decimal comma with a dot, `^` with `**` and the characters × ÷ − with `* / -`. Apart from digits, operators, parentheses, dot, `e`, spaces, `^` and `%`, the text raises `invalid characters in expression`. An empty text raises `empty expression`. Neither has its own `-errorcode`.

Without `%`, every integer becomes `n.0`, so that `10/4` is a floating-point division. With `%` the number stays an integer, because modulo needs integer operands. The computation runs in a

safe interpreter. `Inf`, `-Inf` and `NaN` become `divide by zero`. A result ending in `.0` loses that ending.

`-history 1` shows the previous lines. `-onresult` receives the result.

11.9 In short

- `tkucalendar::getDate` is ISO. `getFormatted` follows `-dateformat`.
- `tkutical` and `tkumonthcanvas` need tical and their own `package require`.
- `tkuical` and `tkutodo` read iCalendar through `tuical`.
- `tkucalc::evaluate` computes in a safe interpreter.

Next chapter: Viewing Data

12 Viewing Data

The widgets of this chapter show what the engines in tclutils already read: CSV, JSON, XML, INI, Markdown, SQLite, diff, hex, strings, Base64, fuzzy search, ZIP and PDF structure. Plus the DAV browser and the text editor. The engine API is in the manual *tclutils*.

`tkuxml`, `tkusqlite` and the editor are loaded separately. The other viewers of this chapter are in the umbrella unless the chapter says otherwise.

12.1 CSV: tkucsv

tkucsv 0.1 shows a table through `tclutils::tucsv`.

```
set w [::tkutils::tkucsv::widget .w ?-height n? ?-header 0|1?]
::tkutils::tkucsv::setData  $w text ?option wert ...?
::tkutils::tkucsv::loadFile $w pfad ?option wert ...?
::tkutils::tkucsv::getRows  $w
```

Options of `setData` and `loadFile` go to `tucsv::parse`. `getRows` returns all rows, the header row included.

12.2 JSON: tkujson

tkujson 0.1 shows the tree of `tujson::parseTyped`. Objects, arrays and scalars stay distinct.

```
set w [::tkutils::tkujson::widget .w ?-height n?]
::tkutils::tkujson::setJson  $w text
::tkutils::tkujson::loadFile $w pfad
::tkutils::tkujson::getTree  $w
```

`setJson` returns the type of the root. `getTree` returns `{type value}`. An object appears as `object (n)`, an array as `array (n)`. Keys keep their order. Array elements are called `[0]`, `[1]` and so on.

12.3 XML: tkuxml

tkuxml 0.1 needs tDOM and is outside the umbrella.

```
set w [::tkutils::tkuxml::widget .w ?-height n?]
::tkutils::tkuxml::setXml   $w text
::tkutils::tkuxml::loadFile $w pfad
::tkutils::tkuxml::getRoot  $w
```

`setXml` returns the name of the root element. The tree shows element names, attributes and text.

12.4 INI: tkuini

tkuini 0.2 shows sections on the left and the pairs of the selected section on the right. The section `""` is called `(global)`. The engine is `tclutils::tuini`.

```
set w [::tkutils::tkuini::widget .w]
::tkutils::tkuini::loadText       $w text
::tkutils::tkuini::loadFile       $w datei
::tkutils::tkuini::sections       $w
::tkutils::tkuini::selectSection  $w abschnitt
::tkutils::tkuini::currentSection $w
::tkutils::tkuini::data           $w
```

`loadText` returns the number of sections.

12.5 Markdown: tkumd and tkumdview

tkumd 0.1 shows the headings as a tree through `tclutils::tumd`.

```
set w [::tkutils::tkumd::widget .w ?-height n?]
::tkutils::tkumd::setMarkdown $w text
::tkutils::tkumd::loadFile    $w pfad
::tkutils::tkumd::getHeadings $w
```

`getHeadings` returns a list `{level title}`.

tkumdview 0.1 shows the same outline and a preview.

```
set w [::tkutils::tkumdview::widget .w ?-width n? ?-height n?]
::tkutils::tkumdview::loadFile    $w pfad
::tkutils::tkumdview::setMarkdown $w text
::tkutils::tkumdview::getMarkdown $w
::tkutils::tkumdview::headings    $w
::tkutils::tkumdview::toHtml      $w
```

`loadFile` reads UTF-8. `loadFile` and `setMarkdown` return the number of characters. `tocWidget` and `textWidget` return the two inner widgets.

12.6 SQLite: tkusqlite

tkusqlite 0.1 needs the package `sqlite3` and is outside the umbrella. The tables are on the left, the rows of the selected table on the right. `tclutils::tusqlite` is the engine without a user interface.

```
set w [::tkutils::tkusqlite::widget .w ?-height n?]
::tkutils::tkusqlite::openFile  $w datei
::tkutils::tkusqlite::tables    $w
::tkutils::tkusqlite::showTable $w name
::tkutils::tkusqlite::getRows   $w
::tkutils::tkusqlite::closeDb   $w
```

`openFile` also takes `:memory:` and returns the number of tables. `showTable` returns the number of rows.

12.7 Diff, Hex, Strings, Base64

tkudiff 0.1 shows a line comparison through `tclutils::tudiff`.

```
set w [::tkutils::tkudiff::widget .w ?-width n? ?-height n?]
::tkutils::tkudiff::setTexts  $w alt neu ?option wert ...?
::tkutils::tkudiff::loadFiles $w alt neu ?option wert ...?
::tkutils::tkudiff::getOps    $w
```

Options go to `tudiff::text`. `getOps` returns a list `{op token}`.

tkuhexedit 0.1 shows and changes bytes through `tuhexdump` and `tubin`.

```
set w [::tkutils::tkuhexedit::widget .hex]
::tkutils::tkuhexedit::loadFile   $w datei
::tkutils::tkuhexedit::gotoOffset $w 0x100
::tkutils::tkuhexedit::findHex    $w {50 4B 03 04}
::tkutils::tkuhexedit::patchHex   $w 0x100 {00 FF}
::tkutils::tkuhexedit::saveFile   $w
```

tkustrings 0.1 lists readable ASCII runs through `tclutils::tustrings`. `setData` and `loadFile` take `-minlength` and return the count. `getStrings` returns the runs.

tkubase64 0.1 encodes and decodes through `tclutils::tubase64`. `encode` and `decode` return the output text. `setInput`, `getInput` and `getOutput` read and write the two fields.

12.8 Fuzzy List: tkufuzzy

tkufuzzy 0.1 filters a list through `tclutils::tufuzzy`.

```
set w [::tkutils::tkufuzzy::widget .w ?-height n?]
::tkutils::tkufuzzy::setItems     $w liste
::tkutils::tkufuzzy::filter       $w muster
::tkutils::tkufuzzy::getMatches   $w
::tkutils::tkufuzzy::getSelection $w
```

`filter` returns the matches by rank and sets the displayed list. `getSelection` returns the selected entry or `""`.

12.9 ZIP: tkuzip

tkuzip 0.1 lists the members of an archive through `tclutils::tuzip`: name, uncompressed size, compressed size, method.

```
set w [::tkutils::tkuzip::widget .w ?-height n?]
::tkutils::tkuzip::openFile       $w archiv
::tkutils::tkuzip::getEntries     $w
::tkutils::tkuzip::selectedMember $w
```

`openFile` returns the number of members. `getEntries` returns the dicts of `tuzip::entries`. `selectedMember` returns the name or `""`.

12.10 PDF Structure: tkupdfinspect

tkupdfinspect 0.1 shows the structure of a PDF file through `tclutils::tupdf`. The text of the pages is a different task (`tupdftext` in the manual *tclutils*).

```
set w [::tkutils::tkupdfinspect::widget .w ?-width n? ?-height n?]
::tkutils::tkupdfinspect::loadFile    $w pfad
::tkutils::tkupdfinspect::currentFile $w
::tkutils::tkupdfinspect::summary     $w
```

`loadFile` fills the tree and returns the path. `summary` is the dict of the last load. `currentFile` is `""` as long as nothing is loaded. `treeWidget` and `textWidget` return tree and detail text. The detail is read-only.

12.11 DAV

tkudavaccount 0.1 is the form for URL, user, password and kind (`carddav`, `caldav`, `webdav`). The test button sends a PROPFIND through `tclutils::tudav` and blocks meanwhile.

```
set a [::tkutils::tkudavaccount::widget .a ?-type carddav? ?-command
    befehl?]
::tkutils::tkudavaccount::setConfig $a {url https://h/c/ user alice
    password pw}
::tkutils::tkudavaccount::testConnection $a
```

`testConnection` returns `1` or `0`. `-command` receives `ok` and a message.

tkudavbrowser 0.2 lists the collections of a CalDAV or CardDAV server, grouped by kind. It is read-only, except with `-editable 1`.

```
set b [::tkutils::tkudavbrowser::widget .b ?-oncollection befehl?]
::tkutils::tkudavbrowser::setClient .b $client
::tkutils::tkudavbrowser::refresh    .b
::tkutils::tkudavbrowser::setData    .b $sammlungen
```

`refresh` calls `discover` and `listCollections` and blocks. The same href appears once, even when a server returns both kinds from one home set. `setData` builds the tree from ready-made dicts `{href displayname kind}` without network access. `-oncollection` receives the widget and the dict of the selection. `collections` returns the flat list, `selected` the dict or `{}`.

12.12 Editor: tkueditor

tkueditor 0.2 is a text editor with a toolbar and a status line. It requires `tkuicon`, `tkutoolbar` and `tkustatus`, plus `tclutils::tuiconv`. The module header calls it a prototype.

```
set w [::tkutils::tkueditor::widget .w ?-width n? ?-height n? ?-wrap
    modus?]
::tkutils::tkueditor::setText    $w text
::tkutils::tkueditor::getText    $w
::tkutils::tkueditor::loadFile   $w pfad ?-encoding name?
::tkutils::tkueditor::saveFile   $w ?pfad? ?-encoding name? ?-eol
    lf|crlf?
::tkutils::tkueditor::find       $w nadel ?-from index? ?-nocase?
::tkutils::tkueditor::replace    $w nadel ersatz ?-all? ?-nocase?
::tkutils::tkueditor::gotoLine   $w n
::tkutils::tkueditor::isModified $w
```

`setText` clears the "modified" flag. `find` returns an index or `""`. `replace` returns the count. `readonly` locks input; `setText` remains possible. `statusbarWidget` returns the `tkustatus` bar, or `""` when it is switched off.

12.13 In short

- The viewers call the engine of the same name in tclutils. XML needs tDOM, SQLite the package `sqlite3`.
- `tkujson::setJson` returns the type of the root, `tkuxml::setXml` the name of the root element.
- `tkudavbrowser::setData` builds the tree without network access. `refresh` blocks.
- `tkueditor` 0.2 is marked as a prototype.

Next chapter: Designer and load

13 Designer and load

13.1 tkurender / tkuload

tkurender is the rendering core of tkudesigner and tkuload; it knows the widget type placeholder (options `text`, `role`, `width`, `height`). tkuload builds a live user interface from a `.tkd` design and offers slot: empty a placeholder frame, and the host packs the real widget into it.

Both modules have namespaces of their own (`::tkurender`, `::tkuload`) and are not in the umbrella. The packages are called `tkutils::tkurender` (0.2) and `tkutils::tkuload` (0.3).

```
package require tkutils::tkuload
package require tkutils::tkurender
package require tkutils::tkufilelist

# Entwurf mit einem benannten Platzhalter (sonst aus tkudesigner, .tkd)
```

```
::tkurender::newModel
set p [::tkurender::addNode placeholder $::tkurender::D(root)]
::tkurender::setName $p filelist
::tkurender::setOpt  $p role filelist
set spec [::tkurender::serialize]

ttk::frame .host
pack .host -fill both -expand 1
set ui [::tkuload::build .host $spec]

set box [::tkuload::slot $ui filelist]
::tkutils::tkufilelist::widget $box.l -dir [pwd]
pack $box.l -fill both -expand 1
pack propagate $box 1
```

From a file: `::tkuload::buildFromFile $parent $pfad`.

`build` returns a dict with `root`, `byId`, `byType` and `byName`.. is also allowed as `$parent`; `build` then creates a frame of its own, `.tkuloadHost<n>`. Build every design into its own empty parent: the widget names come from the node ids, so a second build into the same parent collides. On the first `build`, `tkuload` adds the paths from `TCLUTILS_TM` and `TKUTILS_TM` to the module path, if set.

Further commands for named widgets: `widgetByName`, `setValue`, `getValue`, `fill` (dict name → value, unknown names are skipped) and `collect` (values back as a dict). Since 0.3 `namespace export` names all eight commands, so `namespace import ::tkuload::*` brings in all of them; up to 0.2 only `build`, `buildFromFile` and `slot` were exported.

13.2 Fixed Size of the Placeholder

The placeholder is a `ttk::frame` with a fixed size — by default **200×120** pixels — and `pack propagate 0`. `slot` removes only the children; size and propagation stay. A widget packed into it is therefore clipped to 200×120. The frame from the example is exactly 200×120 with propagation 0 before `pack propagate $box 1`, and 199×251 afterwards (Tcl/Tk 9.0.4: 199×231). Remedies:

- `pack propagate $box 1` — the frame grows with its content, or
- set `width`/`height` suitably in the designer, or
- stretch the placeholder in the layout with `-fill both -expand 1` (or a grid weight).

13.3 Use

According to the book *Utils-Stack*, the app **Dateiblick** stays hand-wired (no `.tkd`); its code is not in tclutils, tkutils or ctrlutils and is not checked here. tkudesigner shows placeholders in the palette (group *Widget*; in the code `defType placeholder Widget`).

Errors of tkuload:

Code	Message
{TKULOAD PARENT}	no such parent widget: .nix

Code	Message
{TKULOAD NONAME}	no widget named 'foo' (from slot and widgetByName)

Details: `docs/tkuload.md` in the tkutils tree.

13.4 Page Layout: tkulayoutcanvas

tkulayoutcanvas 0.1 places named blocks on a page in millimeters. The engine is `tclutils::tulayout`. Fine typography and columns stay in the application.

```
::tkutils::tkulayoutcanvas::widget     $pfad ?option wert ...?
::tkutils::tkulayoutcanvas::getBlocks  $pfad
::tkutils::tkulayoutcanvas::setBlocks  $pfad $blöcke
::tkutils::tkulayoutcanvas::select     $pfad $id
::tkutils::tkulayoutcanvas::getCanvas  $pfad
::tkutils::tkulayoutcanvas::redraw     $pfad
```

Option	Default	Effect
-paper	a4	a4, a5, letter
-orientation	portrait	portrait or landscape
-margin	20.0	margin in mm
-gridmm	5.0	grid in mm
-scale	3.0	pixels per mm
-blocks	{}	dict of the blocks
-definitions	{}	catalog: label and size per id

`mergePreset` takes only the named keys from a preset, by default `x`, `y`, `w` and `show`.

13.5 Summary

- placeholder/slot for host apps and the designer.
- Placeholder 200×120, `pack propagate 0` — keep it in mind when filling.
- `build` returns `root`, `byId`, `byType`, `byName`; values through `setValue/getValue/fill/collect`.
- Required for Dateiblick: no.
- `tkulayoutcanvas` places blocks in millimeters. The engine is `tulayout`.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: the example with `slot`, the sizes of the placeholder, the errors of `tkuload`.

Not tested: the app Dateiblick.

Next chapter: Images

14 Images

Loading, scaling and writing to a file. `tkuimage`, `tkuwinico` and `tkudhash` are in the umbrella. `tkuicon` and `tkucanvaspng` come separately: `tkuicon` needs an extension for SVG, `tkucanvaspng` the engine `tclutils::tupngdraw`.

14.1 Viewing: tkuimage

tkuimage 0.2 provides helpers and a viewer with scrolling and zoom. Scaling uses `imgtools` when it is loaded, otherwise `subsample` and `zoom` of the Tk photo. `imgtools` is optional.

```
::tkutils::tkuimage::fit      bildB bildH kastenB kastenH
    ?contain|cover|none?
::tkutils::tkuimage::load     datei ?ziel?
::tkutils::tkuimage::fromData bytes ?ziel?
::tkutils::tkuimage::scale    quelle b h ?ziel?
::tkutils::tkuimage::thumbnail quelle max ?ziel?
```

`fit` returns `{newW newH scale}`. `load` and `fromData` know PNG and GIF from Tk itself. `contain` fits the image into the box, `cover` fills it.

```
set v [::tkutils::tkuimage::view .v ?-fitmode contain? ?-onchange
    befehl?]
::tkutils::tkuimage::openFile  .v datei.png
::tkutils::tkuimage::zoomIn    .v
::tkutils::tkuimage::zoomOut   .v
::tkutils::tkuimage::zoom1     .v
::tkutils::tkuimage::fitView   .v
::tkutils::tkuimage::getImage  .v
::tkutils::tkuimage::zoomLevel .v
```

`zoomLevel` is `1.0` at 100 percent. `getImage` is the photo at its original size.

14.2 Icons: tkuicon

tkuicon 0.1 loads SVG and PNG and creates a fixed set from `tclutils::tusvg`. Results stay in the cache. SVG under Tk 8.6 needs `tksvg`. Tk 9 draws SVG itself. Without SVG, files still load PNG and GIF. `create` then raises `{TKUTILS TKUICON NOSVG}`.

```
::tkutils::tkuicon::hassvg
::tkutils::tkuicon::available
::tkutils::tkuicon::create name größe ?-color farbe? ?-strokeWidth n?
    ?-name bild?
::tkutils::tkuicon::load   datei ?-height h? ?-width b? ?-name bild?
::tkutils::tkuicon::rescale bildname neueHöhe
::tkutils::tkuicon::loadToolbarSet größe ?-color farbe? ?-icons {namen
    ...}?
::tkutils::tkuicon::clearCache
```

`create` returns the name of the photo. `loadToolbarSet` returns a dict. `rescale` is named so because `scale` is a Tk command. With SVG it redraws, otherwise it scales the existing image. The toolbar that takes the photo as `-icon` is in the chapter Scaffolding.

14.3 Canvas as PNG: tkucanvaspng

tkucanvaspng 0.2 writes a Tk canvas through `tclutils::tupngdraw`. Real text needs glyphs and `-fontmap`.

```
set png [::tkutils::tkucanvaspng::render $canvas \
    ?-region {x1 y1 x2 y2}? ?-scale n? ?-background farbe? \
    ?-textcmd befehl? ?-fontmap dict?]
::tkutils::tkucanvaspng::write aus.png $canvas ?dieselben Optionen?
```

The default for `-region` is `[$canvas bbox all]`. An empty canvas raises `{TKUTILS TKCANVASPNG EMPTY}` with `canvas is empty`. A window that does not exist raises `{TKUTILS TKCANVASPNG WINDOW}` with `not a canvas window: .nix`. An existing widget that is not a canvas is not checked by the module; it fails with the Tk error `bad command "bbox"`. An unknown option reports `{TCLUTILS COMMON OPTION <opt>}`. `-scale` is a positive integer. `-background` is the background of the canvas, otherwise white. `write` returns the file name.

14.4 Windows Icon: tkuwinico

tkuwinico 0.1 lets Tk draw the individual sizes and hands the PNG data to `tclutils::tuico`. Transparency is kept because the payload is PNG.

```
::tkutils::tkuwinico::fromSvg    logo.svg app.ico
::tkutils::tkuwinico::fromPhoto  $photo app.ico ?-sizes {32 16}?
::tkutils::tkuwinico::fromPhotos {{32 $groß} {16 $klein}} app.ico
```

`fromSvg` draws every size from the source. `fromPhoto` scales by integer factors. `fromPhotos` takes a finished image per size and does not scale. SVG output needs the same SVG support as `tkuicon`.

14.5 Similarity: tkudhash

tkudhash 0.1 loads an image file as a photo and hands the pixels to `tclutils::tudhash`. PNG, GIF and PPM work with Tk alone. JPEG and TIFF need the package `Img`.

```
::tkutils::tkudhash::fromFile     pfad ?-sample n?
::tkutils::tkudhash::similarFiles pfadA pfadB ?abstand?
::tkutils::tkudhash::distance     a b
::tkutils::tkudhash::similar      a b ?abstand?
```

`fromFile` returns 16 hex characters, the 64-bit hash. `-sample` is the edge length of the sample. `distance` and `similar` come from `tudhash`. `distance` lies between 0 and 64. `similarFiles` and `similar` use a default distance of 10.

14.6 In short

- `tkuimage::fit` returns width, height and scale. `imgtools` is optional.
- `tkuicon` and `tkuwinico` need `tksvg` or Tk 9 for SVG.

- `tkucanvaspng` writes PNG through `tupngdraw`. An empty canvas raises `EMPTY`.
- `tkudhash` does not compute itself. The hash comes from `tudhash`.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: loading the modules, `tkuimage::fit`, the errors of `tkucanvaspng` and `tkuicon`. Under 8.6.14 without tksvg, `tkuicon::hassvg` returns `0` and `create` the error `NOSVG`; under 9.0.4, `hassvg` returns `1`.

Next chapter: tkdevtools

15 tkdevtools

15.1 The App

tkdevtools (`apps/tkdevtools/tkdevtools.tcl`) is a reference and probe browser for Tk development: reference tables, cheat sheets and probes in one app with a navigation tree. Display only — in normal operation the app writes no files, only the clipboard. The exception is the screenshot mode.

```
wish apps/tkdevtools/tkdevtools.tcl
tclsh apps/tkdevtools/tkdevtools.tcl --shot out.png shadow
```

The screenshot mode renders one tool (keys such as `shadow`, `colors`, `regexp`) into a PNG file of 900×600 pixels and exits. It needs the package `img::window` (Img); if it is missing, the app tries ImageMagick's `import -window root`. Without a key, the first tool (*Tcl / Tk*) stays selected.

With `tclsh` and the keys `shadow`, `colors` and `regexp`, three PNG files of 900×600 pixels are created, exit code 0.

Requirements: Tcl/Tk 8.6 or 9.x, at least **tclutils 0.64.0** (`tucolor`, `tustr` 0.2, `tudate` 0.2, `tupkgfinder` 0.2, `tuappinfo`, `tupath`, `tunum` 0.4; 0.68.0 recommended), from tkutils `tkutablelist` 0.4 and `tkuunits`, plus Tablelist. The app loads `tablelist_tile` before `tkutablelist`; the `tkutl*` helpers use the same variant (chapter Understanding tkutils). The helpers `tkutlsort`, `tkutlfind`, `tkutlclip` are used when present. The app finds the libraries through `apps/_lib/paths.tcl` (the same logic as `tools/setup.tcl`).

The button *Copy report* in the tool *Tcl / Tk* copies an environment report (`tuappinfo::buildReport`) to the clipboard.

15.2 The Tools

Group	Tools
Runtime	Tcl /Tk, OS /platform, Packages, Package shadows, Paths, Display, Environment, Channels, Photo formats, External tools
Reference	Colors, Characters, Graphemes, Fonts, Bitmaps, Font aliases
Conversion / time	Units, Timezones, clock format, Number format
Platform	Paths, glob, Programs, Encoding profiles

Group	Tools
Tk widgets / style	Widget tree, Clipboard, Cursors, Relief /Anchor, Keysym probe, ttk Themes, Widget explorer
Layout	pack, grid
Text / patterns	regexp, format / scan, Encodings, Virtual events

Bold: since 0.46.0. *Paths* exists twice: under *Runtime* the search paths (`auto_path`, `tcl_pkgPath`, `tcl::tm::path`, encoding folders), under *Platform* the path probe.

15.3 Tools Since 0.46.0

Package shadows. Every copy of a package in module paths and `auto_path` (`tupkgfinder::shadows`). Marked red is a second copy of the same version that is never loaded — for example an installation next to the working tree. The row has a light red background, the active copy is written in green, other versions in gray. Which one wins depends on the path order.

Paths (Platform). `file pathtype`, `split`, `dirname`, `normalize`, `nativename` and `tupath::clean` for one path, plus the classification by Windows and Unix rules (`/x` and `C:x` are volume-relative on Windows).

glob. What a pattern matches in a directory, with `-types`. Shows how many dot files `*` leaves out here; on Windows `*` matches them, and `*` plus `.*` returns them twice — the bug that `tkufiletree` fixed (chapter File Widgets).

Programs. `auto_execok` for a name and the PATH entries with matches; separator and, on Windows, PATHEXT.

Encoding profiles. Bytes (hex) read in an encoding with the profiles `strict`, `replace` and `tcl8` (Tcl 9). `strict` stops at the first invalid byte, like a channel with the default profile. Tcl 8.6 has no profiles; the table then shows one row `(8.6)` with the result without an error.

Font aliases. What Tk makes of Helvetica, Times, Courier, Arial and the `Tk*Font` names (family, size, metrics). Yellow: the name comes back unchanged (Tk 9 on Windows for Helvetica) and may coincide in a PDF export with the standard font of the same name. The row then has a yellow background.

Number format. The same values with `format %.Nf` (rounding of the C runtime; glibc and Windows differ at x.5) and `tunum::format` (commercial rounding, 0.5 away from zero), plus the stored double value.

15.4 Summary

- tkdevtools: reference and probes, writes nothing.
- Since 0.46.0: Package shadows, Paths, glob, Programs, Encoding profiles, Font aliases, Number format.
- Needs at least tclutils 0.64.0 (0.68.0 recommended) and Tablelist.
- `--shot` needs Img (`img::window`) or ImageMagick.

Tested with Tcl/Tk 8.6.14 (Img 1.4.16) and 9.0.4 (Img 2.1.1) on Linux under Xvfb: `--shot` with `shadow`, `colors` and `regexp`.

Not tested: display of the encoding profiles under Tcl 9, font aliases on Windows.

Next chapter: Limits

16 Limits

- Without tclutils no sensible pairing (engines, tusettings, tutestrun); at least 0.64.0, 0.68.0 recommended.
- The umbrella needs Tablelist (through `tkufilelist` → `tkutablelist`).
- tkutils loads `tablelist_tile` everywhere; an application that also requires `tablelist` itself fails in tklib.
- Not every module is in the umbrella: `tkuxml`, `tkusqlite`, `tkuicon`, `tkuscrolledframe`, `tkuload`, `tkurender`, the `tkutl*` helpers and others need their own require (chapter Understanding tkutils).
- `tkufiledialog` is not an explorer; `-filetypes` filters only by the first extension of a type.
- tablelist comes from outside (tklib, e.g. `apt install tklib`).
- tical comes from outside; `setup.tcl` does not find it.
- `tkugeometry` does not detect gaps between monitors of different sizes.
- The GUI tests need a display; without one they are skipped.
- Not every module checks its options: `tkutlfooter` skips unknown options.
- Controllers and undo: ctrlutils, not here.
- Checked with Tcl/Tk 8.6.14 and 9.0.4 on Linux (Xvfb). Windows, macOS and the real on-screen appearance are not checked.

16.1 Summary

- View layer; model and controller beside it.
- API reference: module docs `docs/<modul>.md`.
- The gaps of 0.46.0 (Tablelist variants, double click in the file dialog, tests without a display) are fixed in 0.47.0; details in Appendices A and G.

Next chapter: Pitfalls

17 Pitfalls

- Short form `tkudialog::confirm` instead of `::tkutils::tkudialog::confirm` — `invalid command name "tkudialog::confirm"`.
- Setting `TCLUTILS_TM`/`TKUTILS_TM` and leaving out `setup.tcl` — Tcl does not read the variables; `can't find package tkutils` or `can't find package tclutils::…`.
- Umbrella without Tablelist — `can't find package Tablelist`.
- Expecting `tkudialog` 0.2 — file removed, behavior = 0.3.
- Calling `tkudialog::message/warning` — they are called `showInfo`/`showWarning`.
- `input` returns `""` — cancel or empty field, both the same.
- Requiring `tkutablelist` 0.3 and expecting the old PRIMARY selection — it is always 0.4; set `-exportselection 1`.
- `tkufilelist` without `-dir` — shows `/`, not the current directory.

- Calling `tkugeometry::restore` before `track` — redundant; `track` restores by itself.
- `-default 800x600` gives only a size; the window manager chooses the position (under Xvfb without a window manager `+0+0`; not checked under a real window manager). The test `tkugeometry-3.2` therefore compares only the size.
- `setEnabled id 0` (global) after `setEnabled id 0 $w` — clears the window overrides; `invoke` follows only the global flag.
- Filling a placeholder and wondering why the content is clipped — 200×120 and `pack propagate 0`.
- Starting the tests with `wish` — `tclsh tests/all.tcl`.
- Looking for file operations in the widget — they belong in ctrlutils.
- Unknown options to `tkutlfooter` — skipped without an error.
- Copying 0.48.0 over an older tree and leaving the old `.tm` files — then two versions sit side by side (list in the chapter Installation).
- Checking for the error code `{TKUTILS <MODUL> OPTION}` without a fourth element — since 0.47.0 it also carries the option name as the fourth element in `tkuaction` (`define`), `tkuballoon`, `tkuform`, `tkukeynav`, `tkulayoutcanvas`, `tkumarquee`, `tkuscrolledframe`, `tkutablelist`, `tkutoolbar` and `tkuvalidate`: `{TKUTILS <MODUL> OPTION <opt>}`. Up to 0.46.0 it was missing there, and the message was `unknown option '<opt>'` without a `Known:` line. Code that compares the whole code or the message text must be adapted; `lrange $code 0 2` fits both versions. Only a missing value still reports `OPTION` without the option name.

17.1 Fixed in 0.47.0

These pitfalls apply to 0.46.0. They are fixed in 0.47.0:

- Double click on a file in `tkufiledialog` 0.1 — the dialog closes, the result is `""`; up to 0.46.0 *OK* or Return had to be used. Fixed in `tkufiledialog` 0.2.
- Starting the tests without a display — four files abort, exit code 1.
- `tkumonthcanvas.test` without `TICAL_DIR` — all tests skipped when tical lies flat next to tkutils; up to 0.46.0 `TICAL_DIR` had to be set.
- `tkutltools` or a `tkutl*` helper **after** the umbrella — `can't set "usingTile": it is not supported to use both Tablelist and Tablelist_tile in the same application`; up to 0.46.0 `tablelist` had to be loaded first. Now all of them load `tablelist_tile`.

17.2 Summary

- Most pitfalls are questions of paths and loading: setup.tcl, Tablelist, load order.
- Two pitfalls are code bugs in 0.46.0: double click in the file dialog and tests without a display. Both are fixed in 0.47.0.
- Code that detects option errors by their error code must expect the option name as the fourth element in ten more modules since 0.47.0.

Next chapter: Widgets

18 Widgets (Overview)

Overview in the tkutils tree: `docs/guide/tkutils-modules.md`. **75** modules, each with a test, a doc and a man page (`tools/check-docman.tcl` reports no missing files). The umbrella loads 53 directly and `tkutablelist` and `tkuopts` indirectly; the code of the umbrella is authoritative. No parallel versions: `tkutablelist` only 0.4, `tkudialog` only 0.3.

18.1 Excerpt

Widget	Role
tkudialog 0.3	showInfo/Warning/Error, confirm, input, form, build/show
tkufiledialog 0.2	open/save/directory
tkufilelist 0.2 / tkufiletree 0.2	provider UI
tkupath 0.3	path bar, also drive and UNC
tkupreview 0.2	preview, texts through msgcat
tkutablelist 0.4	tablelist wrapper, -exportselection 0
tkugeometry 0.1	remember window geometry
tkuunits 0.1	px/pt/mm/cm/inch
tkuaction 0.2	actions, enable per window
tkutoolbar 0.2	toolbar, addAction
tkulauncher 0.3	program launcher (menu/list)
tkuload 0.3 / tkurender 0.2	UI loading / placeholder (not in the umbrella)
tkuopts 0.1	option check (loaded indirectly)
tkuwinico 0.1	Windows icon from Tk images
tkudhash 0.1	image similarity through tudhash

18.2 Separate Only

Module	Needs
tkuxml 0.1	tDOM
tkusqlite 0.1	sqlite3
tkuicon 0.1	tksvg / Tk 9 for SVG
tkuscrolledframe 0.1	scrollutil
tkutical 0.3, tkumonthcanvas 0.5	tical 1.0
tkucanvaspng 0.2	tclutils::tupngdraw
tkutltools 0.2, tkutl* (0.2/0.3)	Tablelist (tablelist_tile)
tkumdview, tkupdfinspect, tkulayoutcanvas (0.1 each)	tclutils engines

18.3 Summary

- 75 modules: 55 through the umbrella, 20 separately.
- The versions in this appendix are the files in the 0.48.0 tree.

Next chapter: Glossary

19 Glossary

umbrella
Package `tkutils`, loads 53 core modules directly; needs Tablelist.

namespace
`::tkutils::tku<name>` — always call commands fully qualified.

provider
Storage API from tclutils (`tuprovider`); feeds the file widgets.

exportselection
Tk PRIMARY export; off by default in tkutablelist 0.4.

button id
Return value of `tkudialog::show`; `{id Label}` in `-buttons`.

window override
`tkuaction::setEnabled id 0 $w` — enable for one toplevel only.

placeholder
tkurender type: a labeled frame (default 200×120) as a host slot; `slot` empties it.

slot
`::tkuload::slot` — empty a placeholder for a host widget.

tutestrun
Test runner from tclutils; `tclsh tests/all.tcl`.

package shadow
Second copy of a package on the search path; tkdevtools shows it.

pairing
Recommended combination of versions: tclutils 0.68.0 + tkutils 0.48.0 + ctrlutils 0.4.

tablelist_tile
ttk variant of Tablelist; `tkutablelist` and the `tkutl*` helpers load it. The two variants exclude each other.

19.1 Summary

- Short definitions of the terms that recur in the book.

Next chapter: Error Messages

20 Error Messages

Code / symptom	Meaning
invalid command name "tkudialog::..."	namespace missing; ::tkutils::tkudialog::...
can't find package tclutils::...	tclutils not on the module path
can't find package tkutils::...	setup.tcl / tcl::tm::path add missing
can't find package Tablelist	umbrella or table widget without tklib
can't set "usingTile": ...	up to 0.46.0: tkutl* loaded after tkutablelist, load tablelist first; fixed in 0.47.0. Today only when an application requires both variants itself
{TKUTILS <MODUL> OPTION <opt>}	unknown option (modules with tkuopts, tkudialog, tkufiledialog, since 0.47.0 ten more, see below)
{TKUTILS <MODUL> OPTION}	value for the option missing; up to 0.46.0 in ten modules also an unknown option
{TCLUTILS COMMON OPTION <opt>}	unknown option in tkugeometry, tkucanvaspng, tkudavbrowser::refresh
{TKUTILS TKUDIALOG PARENT <w>}	-parent does not exist
{TKUTILS TKUFILEDIALOG PARENT <w>}	the same in the file dialog
{TKUTILS TKUGEOMETRY WINDOW}	no toplevel for save/track/restore
{TCLUTILS TUSETTINGS ...}	geometry persistence (tusettings), e.g. APP
{TKUTILS TKUUNITS UNIT}	unknown unit
{TKUTILS TKUUNITS VALUE}	not a number (2c)
{TKUTILS TKUUNITS WINDOW}	no such window
{TKUTILS TKUACTION WINDOW}	unknown window for enable
{TKUTILS TKUACTION NOACTION} / NOGROUP	unknown action / group
{TKULOAD PARENT} / {TKULOAD NONAME}	tkuload: parent / name missing
unknown interpreter: <key>	tkulauncher: no such interpreter key
ITEM_EXISTS in tkufiletree	before 0.46.0: dot files twice on Windows
{TKUTILS TKCANVASPNG WINDOW} / EMPTY	tkucanvaspng: window missing /canvas empty
{TKUTILS TKUICON NOSVG}	tkuicon: SVG without tksvg under Tk 8.6

The option check of tkuopts is used by 20 modules, among them `tkufilelist`, `tkufiletree`, `tkupath` and `tkulauncher`. They report an unknown option as `{TKUTILS <MODUL> OPTION <opt>}` with `unknown option "<opt>"` and a second line `Known:` with the sorted known options. A missing value reports `{TKUTILS <MODUL> OPTION}` with `missing value for option "<opt>"`. `tkudialog` and `tkufiledialog` check

themselves, with the same structure; there the `Known:` list is in the order of the code.

Ten modules check their options themselves: `tkuaction` (in `define`), `tkuballoon`, `tkuform`, `tkukeynav`, `tkulayoutcanvas`, `tkumarquee`, `tkuscrolledframe`, `tkutoolbar`, `tkuvalidate` and `tkutablelist` (which checks before `tkuopts`). Since 0.47.0 they follow the structure of `tkuopts`: code `{TKUTILS <MODUL> OPTION <opt>}`, message `unknown option "<opt>"` and a second line `Known:` with the sorted known options. Up to 0.46.0 the option name was missing from the code, the `Known:` list was missing, and the message was `unknown option '<opt>'`. Anyone comparing the error code must expect the option as the fourth element since 0.47.0 (Appendix Pitfalls).

Only a few modules use the tclutils check and report `{TCLUTILS COMMON OPTION <opt>}`.

Not every module follows the scheme: `tkutlclip` and `tkutlfind` report `{TKUTILS <MODUL> OPTION}` without the option name and without a `Known:` line, `tkutlstate::autosave` the reason `badoption` in lower case, and `tkutlfooter` does not check options.

Module-specific codes: the respective `docs/<modul>.md`.

20.1 Summary

- Error codes follow `{TKUTILS <MODUL> <GRUND>}`; exceptions are tkuload (`TKULOAD`) and modules with the tclutils check (`TCLUTILS COMMON`).
- Since 0.47.0 the ten modules with their own option check name the option in the code and the known options in the message; up to 0.46.0 both were missing. Modules with the tclutils check report the same under `{TCLUTILS COMMON OPTION <opt>}`. Different: `tkutlclip`, `tkutlfind`, `tkutlstate`, `tkutlfooter`.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: codes and messages of `tkudialog`, `tkutablelist`, `tkutl*`, `tkugeometry`, `tkuunits`, `tkuaction`, `tkuform`, `tkulauncher`, `tkucanvaspng` and `tkuload` as given here and in the chapters. The other rows are read from the code.

21 API catalog: tkutils modules

The catalog lists the procedure signatures of the public commands, in the order of the source file `lib/tm/tkutils/<module>-<version>.tm`. It is generated from the tkutils 0.48.0 sources. Options, return values, error codes and side effects are in the chapters and in `docs/<module>.md`.

Public means: matched by `namespace export`. A signature with `args` takes options or a variable number of arguments; a pair `{name value}` is an argument with a default.

Modules with entries: **74** of 75. `tkutltools` has no commands of its own; it loads the seven `tkutl*` helpers.

21.1 `tkuaction`

Source: `lib/tm/tkutils/tkuaction-0.2.tm`.

```
::tkutils::tkuaction::define {name args}
::tkutils::tkuaction::exists {name}
::tkutils::tkuaction::get {name {key ""}}
::tkutils::tkuaction::names {}
::tkutils::tkuaction::delete {name}
::tkutils::tkuaction::register {name widget}
::tkutils::tkuaction::unregister {name widget}
::tkutils::tkuaction::registerMenu {name menu {index end}}
::tkutils::tkuaction::unregisterMenu {name menu {index ""}}
::tkutils::tkuaction::addMenuItem {menu name}
::tkutils::tkuaction::setEnabled {name enabled {win ""}}
::tkutils::tkuaction::setChecked {name checked}
::tkutils::tkuaction::getEnabled {name {win ""}}
::tkutils::tkuaction::getChecked {name}
::tkutils::tkuaction::toggle {name}
::tkutils::tkuaction::invoke {name}
::tkutils::tkuaction::groupDefine {name actionList}
::tkutils::tkuaction::groupSet {name enabled {win ""}}
::tkutils::tkuaction::groupAdd {name action}
::tkutils::tkuaction::groupList {{name ""}}
::tkutils::tkuaction::reset {}
```

21.2 tkuballoon

Source: `lib/tm/tkutils/tkuballoon-0.1.tm`.

```
::tkutils::tkuballoon::configure {args}
::tkutils::tkuballoon::cget {option}
::tkutils::tkuballoon::enable {}
::tkutils::tkuballoon::disable {}
::tkutils::tkuballoon::add {w text args}
::tkutils::tkuballoon::clear {w}
```

21.3 tkubase64

Source: `lib/tm/tkutils/tkubase64-0.1.tm`.

```
::tkutils::tkubase64::widget {path args}
::tkutils::tkubase64::setInput {path text}
::tkutils::tkubase64::getInput {path}
::tkutils::tkubase64::getOutput {path}
::tkutils::tkubase64::encode {path}
::tkutils::tkubase64::decode {path}
```

21.4 tkubind

Source: `lib/tm/tkutils/tkubind-0.1.tm`.

```
::tkutils::tkubind::platform {{key ""}}
::tkutils::tkubind::accelerator {spec}
::tkutils::tkubind::isEditing {}
::tkutils::tkubind::key {spec script args}
::tkutils::tkubind::unbind {id}
::tkutils::tkubind::context {widget script}
::tkutils::tkubind::doubleClick {widget script}
::tkutils::tkubind::group {cmd args}
::tkutils::tkubind::getInfo {args}
::tkutils::tkubind::clear {}
```

21.5 tkucal

Source: `lib/tm/tkutils/tkucal-0.1.tm`.

```
::tkutils::tkucal::widget {path args}
::tkutils::tkucal::setMonth {path year month}
::tkutils::tkucal::getMonth {path}
::tkutils::tkucal::next {path}
::tkutils::tkucal::prev {path}
::tkutils::tkucal::today {path}
::tkutils::tkucal::getText {path}
```

21.6 tkucalc

Source: `lib/tm/tkutils/tkucalc-0.2.tm`.

```
::tkutils::tkucalc::evaluate {text}
::tkutils::tkucalc::widget {path args}
::tkutils::tkucalc::getHistory {path}
::tkutils::tkucalc::clearHistory {path}
::tkutils::tkucalc::clear {path}
```

21.7 tkucalendar

Source: `lib/tm/tkutils/tkucalendar-0.2.tm`.

```
::tkutils::tkucalendar::widget {path args}
::tkutils::tkucalendar::getDate {path}
::tkutils::tkucalendar::getFormatted {path}
::tkutils::tkucalendar::setDate {path iso}
::tkutils::tkucalendar::today {path}
::tkutils::tkucalendar::next {path}
::tkutils::tkucalendar::prev {path}
::tkutils::tkucalendar::getMonth {path}
```

21.8 tkucanvaspng

Source: `lib/tm/tkutils/tkucanvaspng-0.2.tm`.

```
::tkutils::tkucanvaspng::render {canvas args}
::tkutils::tkucanvaspng::write {file canvas args}
```

21.9 tkucontextmenu

Source: lib/tm/tkutils/tkucontextmenu-0.1.tm.

```
::tkutils::tkucontextmenu::create {path args}
::tkutils::tkucontextmenu::addItem {menu label args}
::tkutils::tkucontextmenu::addSeparator {menu}
::tkutils::tkucontextmenu::addSubmenu {menu label submenu}
::tkutils::tkucontextmenu::addCheckItem {menu label args}
::tkutils::tkucontextmenu::addRadioItem {menu label args}
::tkutils::tkucontextmenu::attach {menu widget args}
::tkutils::tkucontextmenu::detach {menu widget}
::tkutils::tkucontextmenu::show {menu x y}
::tkutils::tkucontextmenu::enable {menu label}
::tkutils::tkucontextmenu::disable {menu label}
::tkutils::tkucontextmenu::setCallback {menu label command}
::tkutils::tkucontextmenu::setUpdateHandler {menu handler}
::tkutils::tkucontextmenu::destroyMenu {menu}
::tkutils::tkucontextmenu::createStandardEdit {path args}
::tkutils::tkucontextmenu::createFromSpec {path spec}
```

21.10 tkucsv

Source: lib/tm/tkutils/tkucsv-0.1.tm.

```
::tkutils::tkucsv::widget {path args}
::tkutils::tkucsv::setData {path csvText args}
::tkutils::tkucsv::loadFile {path filename args}
::tkutils::tkucsv::getRows {path}
```

21.11 tkudateentry

Source: lib/tm/tkutils/tkudateentry-0.1.tm.

```
::tkutils::tkudateentry::widget {path args}
::tkutils::tkudateentry::getText {path}
::tkutils::tkudateentry::getDate {path}
::tkutils::tkudateentry::setDate {path iso}
::tkutils::tkudateentry::today {path}
::tkutils::tkudateentry::clear {path}
```

21.12 tkudavaccount

Source: lib/tm/tkutils/tkudavaccount-0.1.tm.

```
::tkutils::tkudavaccount::widget {path args}
::tkutils::tkudavaccount::getConfig {path}
::tkutils::tkudavaccount::setConfig {path cfg}
::tkutils::tkudavaccount::clear {path}
::tkutils::tkudavaccount::focusUrl {path}
::tkutils::tkudavaccount::testConnection {path}
```

21.13 tkudavbrowser

Source: `lib/tm/tkutils/tkudavbrowser-0.2.tm`.

```
::tkutils::tkudavbrowser::widget {path args}
::tkutils::tkudavbrowser::treeWidget {path}
::tkutils::tkudavbrowser::setClient {path client}
::tkutils::tkudavbrowser::refresh {path args}
::tkutils::tkudavbrowser::setData {path cols}
::tkutils::tkudavbrowser::collections {path}
::tkutils::tkudavbrowser::selected {path}
::tkutils::tkudavbrowser::createCalendar {path displayname}
::tkutils::tkudavbrowser::createAddressbook {path displayname}
::tkutils::tkudavbrowser::deleteCollection {path {href ""}}
```

21.14 tkudhash

Source: `lib/tm/tkutils/tkudhash-0.1.tm`.

```
::tkutils::tkudhash::fromFile {path args}
::tkutils::tkudhash::distance {a b}
::tkutils::tkudhash::similar {a b {maxDist 10}}
::tkutils::tkudhash::similarFiles {pathA pathB {maxDist 10}}
```

21.15 tkudialog

Source: `lib/tm/tkutils/tkudialog-0.3.tm`.

```
::tkutils::tkudialog::choose {win id}
::tkutils::tkudialog::build {win args}
::tkutils::tkudialog::getText {win}
::tkutils::tkudialog::getDetail {win}
::tkutils::tkudialog::result {win}
::tkutils::tkudialog::copyText {win}
::tkutils::tkudialog::show {win args}
::tkutils::tkudialog::showInfo {message args}
::tkutils::tkudialog::showWarning {message args}
::tkutils::tkudialog::showError {message args}
::tkutils::tkudialog::confirm {message args}
::tkutils::tkudialog::input {args}
::tkutils::tkudialog::form {fieldspec args}
```

21.16 tkudiff

Source: lib/tm/tkutils/tkudiff-0.1.tm.

```
::tkutils::tkudiff::widget {path args}
::tkutils::tkudiff::setTexts {path oldText newText args}
::tkutils::tkudiff::loadFiles {path oldFile newFile args}
::tkutils::tkudiff::getOps {path}
```

21.17 tkueditor

Source: lib/tm/tkutils/tkueditor-0.2.tm.

```
::tkutils::tkueditor::widget {path args}
::tkutils::tkueditor::toolbarWidget {path}
::tkutils::tkueditor::statusbarWidget {path}
::tkutils::tkueditor::setStatus {path text}
::tkutils::tkueditor::refreshStatus {path}
::tkutils::tkueditor::setText {path text}
::tkutils::tkueditor::getText {path}
::tkutils::tkueditor::loadFile {path filename args}
::tkutils::tkueditor::saveFile {path args}
::tkutils::tkueditor::currentFile {path}
::tkutils::tkueditor::encoding {path args}
::tkutils::tkueditor::eol {path args}
::tkutils::tkueditor::find {path needle args}
::tkutils::tkueditor::findAll {path needle args}
::tkutils::tkueditor::findNext {path needle args}
::tkutils::tkueditor::replace {path needle repl args}
::tkutils::tkueditor::highlightAll {path needle args}
::tkutils::tkueditor::clearHighlight {path args}
::tkutils::tkueditor::gotoLine {path n}
::tkutils::tkueditor::cursor {path}
::tkutils::tkueditor::readonly {path args}
::tkutils::tkueditor::isModified {path}
::tkutils::tkueditor::menuWidget {path}
::tkutils::tkueditor::addMenuItem {path label command}
::tkutils::tkueditor::addMenuSeparator {path}
::tkutils::tkueditor::selectAll {path}
```

21.18 tkufiledialog

Source: lib/tm/tkutils/tkufiledialog-0.2.tm.

```
::tkutils::tkufiledialog::open {args}
::tkutils::tkufiledialog::save {args}
::tkutils::tkufiledialog::directory {args}
```

21.19 tkufilelist

Source: lib/tm/tkutils/tkufilelist-0.2.tm.

```
::tkutils::tkufilelist::widget {path args}
::tkutils::tkufilelist::setDir {path dir}
::tkutils::tkufilelist::refresh {path}
::tkutils::tkufilelist::setFilter {path glob}
::tkutils::tkufilelist::dir {path}
::tkutils::tkufilelist::selectedEntry {path}
::tkutils::tkufilelist::selectedEntries {path}
```

21.20 tkufiletree

Source: `lib/tm/tkutils/tkufiletree-0.2.tm`.

```
::tkutils::tkufiletree::widget {path args}
::tkutils::tkufiletree::setRoot {path dir}
::tkutils::tkufiletree::refresh {path}
::tkutils::tkufiletree::refreshDir {path dir}
::tkutils::tkufiletree::up {path}
::tkutils::tkufiletree::selectedPath {path}
::tkutils::tkufiletree::treeview {path}
::tkutils::tkufiletree::root {path}
::tkutils::tkufiletree::reveal {path target}
```

21.21 tkufilterbar

Source: `lib/tm/tkutils/tkufilterbar-0.2.tm`.

```
::tkutils::tkufilterbar::widget {path args}
::tkutils::tkufilterbar::setColumns {path cols}
::tkutils::tkufilterbar::columns {path}
::tkutils::tkufilterbar::getFilters {path}
::tkutils::tkufilterbar::getFilter {path col}
::tkutils::tkufilterbar::setFilter {path col text}
::tkutils::tkufilterbar::clear {path}
::tkutils::tkufilterbar::focusFirst {path}
```

21.22 tkuform

Source: `lib/tm/tkutils/tkuform-0.1.tm`.

```
::tkutils::tkuform::widget {path fieldspec args}
::tkutils::tkuform::get {path name}
::tkutils::tkuform::setField {path name value}
::tkutils::tkuform::fieldNames {path}
::tkutils::tkuform::values {path}
::tkutils::tkuform::setValues {path dict}
::tkutils::tkuform::widgetOf {path name}
```

21.23 tkufuzzy

Source: `lib/tm/tkutils/tkufuzzy-0.1.tm`.

```
::tkutils::tkufuzzy::widget {path args}
::tkutils::tkufuzzy::setItems {path items}
::tkutils::tkufuzzy::filter {path pattern}
::tkutils::tkufuzzy::getMatches {path}
::tkutils::tkufuzzy::getSelection {path}
```

21.24 tkugeometry

Source: `lib/tm/tkutils/tkugeometry-0.1.tm`.

```
::tkutils::tkugeometry::key {w}
::tkutils::tkugeometry::save {w app args}
::tkutils::tkugeometry::restore {w app args}
::tkutils::tkugeometry::track {w app args}
```

21.25 tkuhexedit

Source: `lib/tm/tkutils/tkuhexedit-0.1.tm`.

```
::tkutils::tkuhexedit::widget {path args}
::tkutils::tkuhexedit::setData {path data}
::tkutils::tkuhexedit::getData {path}
::tkutils::tkuhexedit::loadFile {path filename}
::tkutils::tkuhexedit::saveFile {path {filename ""}}
::tkutils::tkuhexedit::render {path {startOffset 0}}
::tkutils::tkuhexedit::gotoOffset {path offset}
::tkutils::tkuhexedit::findText {path pattern {start 0}}
::tkutils::tkuhexedit::findHex {path hex {start 0}}
::tkutils::tkuhexedit::patchHex {path offset hex}
```

21.26 tkuical

Source: `lib/tm/tkutils/tkuical-0.2.tm`.

```
::tkutils::tkuical::widget {path args}
::tkutils::tkuical::treeWidget {path}
::tkutils::tkuical::loadText {path ics}
::tkutils::tkuical::loadFile {path file}
::tkutils::tkuical::setComponents {path comps}
::tkutils::tkuical::events {path}
::tkutils::tkuical::count {path}
::tkutils::tkuical::addEvent {path {summary ""}}
::tkutils::tkuical::removeEvent {path index}
::tkutils::tkuical::setField {path index field value}
::tkutils::tkuical::setEventProperty {path index name value {params
    {}}}
::tkutils::tkuical::addEventProperty {path index name value {params
    {}}}
::tkutils::tkuical::removeEventProperty {path index name}
::tkutils::tkuical::toText {path}
::tkutils::tkuical::save {path file}
```

21.27 tkuicon

Source: `lib/tm/tkutils/tkuicon-0.1.tm`.

```
::tkutils::tkuicon::hassvg {}
::tkutils::tkuicon::load {filename args}
::tkutils::tkuicon::create {iconName size args}
::tkutils::tkuicon::rescale {imgName newHeight}
::tkutils::tkuicon::available {}
::tkutils::tkuicon::clearCache {}
::tkutils::tkuicon::loadToolbarSet {size args}
```

21.28 tkuimage

Source: `lib/tm/tkutils/tkuimage-0.2.tm`.

```
::tkutils::tkuimage::fit {imgW imgH boxW boxH {mode contain}}
::tkutils::tkuimage::load {file {dst ""}}
::tkutils::tkuimage::fromData {bytes {dst ""}}
::tkutils::tkuimage::scale {src w h {dst ""}}
::tkutils::tkuimage::thumbnail {src maxSize {dst ""}}
::tkutils::tkuimage::view {path args}
::tkutils::tkuimage::openFile {path file}
::tkutils::tkuimage::getImage {path}
::tkutils::tkuimage::zoomLevel {path}
::tkutils::tkuimage::zoomIn {path}
::tkutils::tkuimage::zoomOut {path}
::tkutils::tkuimage::zoom1 {path}
::tkutils::tkuimage::fitView {path}
```

21.29 tkuini

Source: lib/tm/tkutils/tkuini-0.2.tm.

```
::tkutils::tkuini::widget {path args}
::tkutils::tkuini::sectionWidget {path}
::tkutils::tkuini::treeWidget {path}
::tkutils::tkuini::loadText {path ini}
::tkutils::tkuini::loadFile {path file}
::tkutils::tkuini::setData {path d}
::tkutils::tkuini::selectSection {path section}
::tkutils::tkuini::data {path}
::tkutils::tkuini::sections {path}
::tkutils::tkuini::currentSection {path}
::tkutils::tkuini::count {path}
::tkutils::tkuini::setKey {path section key value}
::tkutils::tkuini::removeKey {path section key}
::tkutils::tkuini::addSection {path section}
::tkutils::tkuini::removeSection {path section}
::tkutils::tkuini::toText {path}
::tkutils::tkuini::save {path file}
```

21.30 tkujson

Source: lib/tm/tkutils/tkujson-0.1.tm.

```
::tkutils::tkujson::widget {path args}
::tkutils::tkujson::setJson {path jsonText}
::tkutils::tkujson::loadFile {path filename}
::tkutils::tkujson::getTree {path}
```

21.31 tkukeynav

Source: lib/tm/tkutils/tkukeynav-0.1.tm.

```
::tkutils::tkukeynav::next {w}
::tkutils::tkukeynav::prev {w}
::tkutils::tkukeynav::enable {w args}
::tkutils::tkukeynav::disable {w}
::tkutils::tkukeynav::form {container args}
```

21.32 tkulabeled

Source: lib/tm/tkutils/tkulabeled-0.1.tm.

```
::tkutils::tkulabeled::add {path type args}
::tkutils::tkulabeled::value {path args}
::tkutils::tkulabeled::control {path}
::tkutils::tkulabeled::labelwidget {path}
::tkutils::tkulabeled::configure {path args}
```

21.33 tkulauncher

Source: lib/tm/tkutils/tkulauncher-0.3.tm.

```
::tkutils::tkulauncher::widget {path args}
::tkutils::tkulauncher::available {entry}
::tkutils::tkulauncher::resolve {entry}
::tkutils::tkulauncher::load {file}
::tkutils::tkulauncher::reload {path}
::tkutils::tkulauncher::editConfig {path}
::tkutils::tkulauncher::openLocation {path}
::tkutils::tkulauncher::saveEntries {items file}
::tkutils::tkulauncher::showCalendar {path}
::tkutils::tkulauncher::showCalculator {path}
::tkutils::tkulauncher::editEntries {path}
::tkutils::tkulauncher::systemTitle {id}
::tkutils::tkulauncher::resolveSystem {id args}
::tkutils::tkulauncher::systemItems {args}
::tkutils::tkulauncher::systemIds {}
```

21.34 tkulayoutcanvas

Source: `lib/tm/tkutils/tkulayoutcanvas-0.1.tm`.

```
::tkutils::tkulayoutcanvas::widget {path args}
::tkutils::tkulayoutcanvas::configure {path args}
::tkutils::tkulayoutcanvas::cget {path opt}
::tkutils::tkulayoutcanvas::getCanvas {path}
::tkutils::tkulayoutcanvas::getBlocks {path}
::tkutils::tkulayoutcanvas::setBlocks {path blocks}
::tkutils::tkulayoutcanvas::select {path id}
::tkutils::tkulayoutcanvas::mergePreset {path preset args}
::tkutils::tkulayoutcanvas::redraw {path}
```

21.35 tkuldif

Source: `lib/tm/tkutils/tkuldif-0.2.tm`.

```
::tkutils::tkuldif::widget {path args}
::tkutils::tkuldif::treeWidget {path}
::tkutils::tkuldif::loadText {path ldif}
::tkutils::tkuldif::loadFile {path file}
::tkutils::tkuldif::setEntries {path entries}
::tkutils::tkuldif::entries {path}
::tkutils::tkuldif::count {path}
::tkutils::tkuldif::addEntry {path {dn "cn=New"}}
::tkutils::tkuldif::removeEntry {path index}
::tkutils::tkuldif::addAttr {path entryIndex attr value}
::tkutils::tkuldif::setAttr {path entryIndex pairIndex attr value}
::tkutils::tkuldif::removeAttr {path entryIndex pairIndex}
::tkutils::tkuldif::toText {path}
::tkutils::tkuldif::save {path file}
```

21.36 tkuload

Source: lib/tm/tkutils/tkuload-0.3.tm.

```
::tkuload::build {parent spec}
::tkuload::buildFromFile {parent path}
::tkuload::slot {ui name}
::tkuload::widgetByName {ui name}
::tkuload::setValue {ui name value}
::tkuload::getValue {ui name}
::tkuload::fill {ui data}
::tkuload::collect {ui {names ""}}
```

21.37 tkumarquee

Source: lib/tm/tkutils/tkumarquee-0.1.tm.

```
::tkutils::tkumarquee::enable {c args}
::tkutils::tkumarquee::disable {c}
::tkutils::tkumarquee::region {c}
::tkutils::tkumarquee::active {c}
```

21.38 tkumd

Source: lib/tm/tkutils/tkumd-0.1.tm.

```
::tkutils::tkumd::widget {path args}
::tkutils::tkumd::setMarkdown {path markdown}
::tkutils::tkumd::loadFile {path filename}
::tkutils::tkumd::getHeadings {path}
```

21.39 tkumdview

Source: lib/tm/tkutils/tkumdview-0.1.tm.

```
::tkutils::tkumdview::widget {path args}
::tkutils::tkumdview::tocWidget {path}
::tkutils::tkumdview::textWidget {path}
::tkutils::tkumdview::getMarkdown {path}
::tkutils::tkumdview::headings {path}
::tkutils::tkumdview::toHtml {path}
::tkutils::tkumdview::loadFile {path filename}
::tkutils::tkumdview::setMarkdown {path md}
```

21.40 tkumonthcanvas

Source: lib/tm/tkutils/tkumonthcanvas-0.5.tm.

21.41 tkunotes

Source: lib/tm/tkutils/tkunotes-0.2.tm.

```
::tkutils::tkunotes::widget {path args}
::tkutils::tkunotes::store {path}
::tkutils::tkunotes::count {path}
::tkutils::tkunotes::current {path}
::tkutils::tkunotes::treeWidget {path}
::tkutils::tkunotes::refresh {path}
::tkutils::tkunotes::select {path id}
::tkutils::tkunotes::addRoot {path title content {tags {}}}
::tkutils::tkunotes::addChild {path parentId title content {tags {}}}
::tkutils::tkunotes::delete {path id {cascade 1}}
::tkutils::tkunotes::search {path query}
::tkutils::tkunotes::move {path id newParentId}
::tkutils::tkunotes::load {path file}
::tkutils::tkunotes::save {path {file ""}}
::tkutils::tkunotes::expandAll {path}
::tkutils::tkunotes::collapseAll {path}
::tkutils::tkunotes::tags {path}
::tkutils::tkunotes::byTag {path tag}
::tkutils::tkunotes::addTag {path id tag}
::tkutils::tkunotes::removeTag {path id tag}
::tkutils::tkunotes::subtree {path id}
::tkutils::tkunotes::saveSubtree {path id file}
```

21.42 tkunumentry

Source: `lib/tm/tkutils/tkunumentry-0.1.tm`.

```
::tkutils::tkunumentry::widget {path args}
::tkutils::tkunumentry::getValue {path}
::tkutils::tkunumentry::setValue {path value}
::tkutils::tkunumentry::clear {path}
```

21.43 tkuopts

Source: `lib/tm/tkutils/tkuopts-0.1.tm`.

```
::tkutils::tkuopts::merge {mod defaults argv}
```

21.44 tkupath

Source: `lib/tm/tkutils/tkupath-0.3.tm`.

```
::tkutils::tkupath::widget {path args}
::tkutils::tkupath::setPath {path p}
::tkutils::tkupath::getPath {path}
```

21.45 tkupdfinspect

Source: `lib/tm/tkutils/tkupdfinspect-0.1.tm`.

```
::tkutils::tkupdfinspect::widget {path args}
::tkutils::tkupdfinspect::treeWidget {path}
::tkutils::tkupdfinspect::textWidget {path}
::tkutils::tkupdfinspect::currentFile {path}
::tkutils::tkupdfinspect::summary {path}
::tkutils::tkupdfinspect::loadFile {path filename}
```

21.46 tkupreview

Source: `lib/tm/tkutils/tkupreview-0.2.tm`.

```
::tkutils::tkupreview::widget {path}
::tkutils::tkupreview::text {path title content}
::tkutils::tkupreview::markdown {path title content}
::tkutils::tkupreview::message {path msg}
::tkutils::tkupreview::html {path title content}
::tkutils::tkupreview::json {path title content}
::tkutils::tkupreview::xml {path title content}
::tkutils::tkupreview::ini {path title content}
::tkutils::tkupreview::hex {path title data}
::tkutils::tkupreview::csv {path title content args}
::tkutils::tkupreview::photo {path title img}
::tkutils::tkupreview::sqlite {path title file}
::tkutils::tkupreview::image {path title file}
::tkutils::tkupreview::pdf {path title file}
::tkutils::tkupreview::clear {path}
::tkutils::tkupreview::kind {path}
```

21.47 tkurender

Source: `lib/tm/tkutils/tkurender-0.2.tm`.

```
::tkurender::isContainer {type}
::tkurender::optspecOf {type}
::tkurender::labelOf {type}
::tkurender::pkgFor {type}
::tkurender::pkgAvail {type}
::tkurender::newModel {}
::tkurender::addNode {type parent {internal 0}}
::tkurender::deleteNode {id}
::tkurender::nodeType {id}
::tkurender::nodeOpt {id o}
::tkurender::setOpt {id o v}
::tkurender::nodeGeom {id k}
::tkurender::setGeom {id k v}
::tkurender::nodeLayout {id}
::tkurender::setLayout {id v}
::tkurender::nodeName {id}
::tkurender::setName {id v}
```

```
::tkurender::setStretch {id which v}
::tkurender::renderChildren {mid tkparent pv}
::tkurender::serialize {}
::tkurender::deserialize {spec}
```

21.48 tkuscrolledframe

Source: `lib/tm/tkutils/tkuscrolledframe-0.1.tm`.

```
::tkutils::tkuscrolledframe::widget {path args}
::tkutils::tkuscrolledframe::content {path}
::tkutils::tkuscrolledframe::scrollableframe {path}
::tkutils::tkuscrolledframe::scrollarea {path}
::tkutils::tkuscrolledframe::configure {path args}
::tkutils::tkuscrolledframe::see {path w {corner nw}}
::tkutils::tkuscrolledframe::seerect {path x1 y1 x2 y2 {corner nw}}
::tkutils::tkuscrolledframe::xview {path args}
::tkutils::tkuscrolledframe::yview {path args}
::tkutils::tkuscrolledframe::autosize {path dimensions}
::tkutils::tkuscrolledframe::autofillx {path boolean}
::tkutils::tkuscrolledframe::autofilly {path boolean}
```

21.49 tkusearchbar

Source: `lib/tm/tkutils/tkusearchbar-0.1.tm`.

```
::tkutils::tkusearchbar::widget {path args}
::tkutils::tkusearchbar::getText {path}
::tkutils::tkusearchbar::setText {path text}
::tkutils::tkusearchbar::clear {path}
::tkutils::tkusearchbar::getFilter {path}
::tkutils::tkusearchbar::setFilter {path value}
::tkutils::tkusearchbar::setFilters {path filters}
::tkutils::tkusearchbar::focusSearch {path}
```

21.50 tkusqlite

Source: `lib/tm/tkutils/tkusqlite-0.1.tm`.

```
::tkutils::tkusqlite::widget {path args}
::tkutils::tkusqlite::openFile {path dbfile}
::tkutils::tkusqlite::tables {path}
::tkutils::tkusqlite::showTable {path table}
::tkutils::tkusqlite::getRows {path}
::tkutils::tkusqlite::closeDb {path}
```

21.51 tkustatus

Source: `lib/tm/tkutils/tkustatus-0.2.tm`.

```
::tkutils::tkustatus::widget {path args}
::tkutils::tkustatus::setText {path text}
::tkutils::tkustatus::getText {path}
::tkutils::tkustatus::addField {path id args}
::tkutils::tkustatus::setField {path id text}
::tkutils::tkustatus::fieldText {path id}
::tkutils::tkustatus::progress {path {value ""}}
::tkutils::tkustatus::flash {path text {ms 2000}}
```

21.52 tkustrings

Source: `lib/tm/tkutils/tkustrings-0.1.tm`.

```
::tkutils::tkustrings::widget {path args}
::tkutils::tkustrings::setData {path data args}
::tkutils::tkustrings::loadFile {path filename args}
::tkutils::tkustrings::getStrings {path}
```

21.53 tkutab

Source: `lib/tm/tkutils/tkutab-0.2.tm`.

```
::tkutils::tkutab::widget {path args}
::tkutils::tkutab::add {path label}
::tkutils::tkutab::close {path f}
::tkutils::tkutab::current {path}
::tkutils::tkutab::setLabel {path f label}
::tkutils::tkutab::tabs {path}
::tkutils::tkutab::count {path}
```

21.54 tkutablelist

Source: `lib/tm/tkutils/tkutablelist-0.4.tm`.

```
::tkutils::tkutablelist::widget {path args}
::tkutils::tkutablelist::footerWidget {path}
::tkutils::tkutablelist::footerSet {path values}
::tkutils::tkutablelist::footerSum {path args}
::tkutils::tkutablelist::tableWidget {path}
::tkutils::tkutablelist::insert {path row}
::tkutils::tkutablelist::setRows {path rows}
::tkutils::tkutablelist::rows {path}
::tkutils::tkutablelist::clear {path}
::tkutils::tkutablelist::size {path}
::tkutils::tkutablelist::getRow {path index}
::tkutils::tkutablelist::setRow {path index row}
::tkutils::tkutablelist::deleteRow {path index}
::tkutils::tkutablelist::cellText {path row col}
::tkutils::tkutablelist::setCell {path row col value}
::tkutils::tkutablelist::setColumns {path titles}
```

```
::tkutils::tkutablelist::columns {path}
::tkutils::tkutablelist::configureColumn {path col args}
::tkutils::tkutablelist::configureRow {path row args}
::tkutils::tkutablelist::configureCell {path cell args}
::tkutils::tkutablelist::insertChild {path parent values {index end}}
::tkutils::tkutablelist::expand {path {row ""}}
::tkutils::tkutablelist::collapse {path {row ""}}
::tkutils::tkutablelist::selection {path}
::tkutils::tkutablelist::selectedRows {path}
::tkutils::tkutablelist::selectRows {path indices}
::tkutils::tkutablelist::sortBy {path col {order -increasing}}
::tkutils::tkutablelist::loadCsv {path csv args}
::tkutils::tkutablelist::fromNotes {path store args}
::tkutils::tkutablelist::editEndCommand {path cmd}
::tkutils::tkutablelist::selectCommand {path args}
::tkutils::tkutablelist::doubleCommand {path args}
::tkutils::tkutablelist::toCsv {path args}
::tkutils::tkutablelist::saveCsv {path file args}
```

21.55 tkutags

Source: `lib/tm/tkutils/tkutags-0.1.tm`.

```
::tkutils::tkutags::widget {path args}
::tkutils::tkutags::getTags {path}
::tkutils::tkutags::setTags {path tags}
::tkutils::tkutags::addTag {path tag}
::tkutils::tkutags::removeTag {path tag}
::tkutils::tkutags::clear {path}
```

21.56 tkutical

Source: `lib/tm/tkutils/tkutical-0.3.tm`.

```
::tkutils::tkutical::widget {path args}
::tkutils::tkutical::refresh {path}
::tkutils::tkutical::setView {path view}
::tkutils::tkutical::getView {path}
::tkutils::tkutical::setDate {path iso}
::tkutils::tkutical::getDate {path}
::tkutils::tkutical::setMonth {path year month}
::tkutils::tkutical::getMonth {path}
::tkutils::tkutical::next {path}
::tkutils::tkutical::prev {path}
::tkutils::tkutical::today {path}
::tkutils::tkutical::selectMode {path mode}
::tkutils::tkutical::getSelection {path}
::tkutils::tkutical::setSelection {path dates}
::tkutils::tkutical::clearSelection {path}
::tkutils::tkutical::canvasWidget {path}
```

21.57 tkutimeentry

Source: `lib/tm/tkutils/tkutimeentry-0.1.tm`.

```
::tkutils::tkutimeentry::widget {path args}
::tkutils::tkutimeentry::getTime {path}
::tkutils::tkutimeentry::setTime {path str}
```

21.58 tkutlclip

Source: `lib/tm/tkutils/tkutlclip-0.2.tm`.

```
::tkutils::tkutlclip::asText {tbl rows args}
::tkutils::tkutlclip::copySelection {tbl args}
::tkutils::tkutlclip::copyAll {tbl args}
::tkutils::tkutlclip::installBindings {tbl args}
```

21.59 tkutlfind

Source: `lib/tm/tkutils/tkutlfind-0.2.tm`.

```
::tkutils::tkutlfind::find {tbl query args}
::tkutils::tkutlfind::matches {tbl}
::tkutils::tkutlfind::next {tbl}
::tkutils::tkutlfind::prev {tbl}
::tkutils::tkutlfind::clear {tbl}
```

21.60 tkutlfmt

Source: `lib/tm/tkutils/tkutlfmt-0.3.tm`.

```
::tkutils::tkutlfmt::column {tbl col type args}
::tkutils::tkutlfmt::columns {tbl spec}
```

21.61 tkutlfooter

Source: lib/tm/tkutils/tkutlfooter-0.3.tm.

```
::tkutils::tkutlfooter::attach {tbl foot args}
::tkutils::tkutlfooter::detach {tbl foot}
::tkutils::tkutlfooter::setvals {foot values}
::tkutils::tkutlfooter::update {tbl foot}
::tkutils::tkutlfooter::autosum {tbl foot args}
```

21.62 tkutlsort

Source: lib/tm/tkutils/tkutlsort-0.2.tm.

```
::tkutils::tkutlsort::column {tbl col type {cmd ""}}
::tkutils::tkutlsort::columns {tbl spec}
```

21.63 tkutlstate

Source: lib/tm/tkutils/tkutlstate-0.3.tm.

```
::tkutils::tkutlstate::save {tbl}
::tkutils::tkutlstate::restore {tbl state}
::tkutils::tkutlstate::saveToFile {tbl path}
::tkutils::tkutlstate::restoreFromFile {tbl path}
::tkutils::tkutlstate::saveVia {tbl key setPrefix}
::tkutils::tkutlstate::restoreVia {tbl key getPrefix}
::tkutils::tkutlstate::autosave {tbl key setPrefix args}
::tkutils::tkutlstate::cancelAutosave {tbl}
```

21.64 tkutltree

Source: lib/tm/tkutils/tkutltree-0.2.tm.

```
::tkutils::tkutltree::fromData {tbl data args}
::tkutils::tkutltree::toData {tbl args}
::tkutils::tkutltree::clear {tbl args}
```

21.65 tkutodo

Source: lib/tm/tkutils/tkutodo-0.2.tm.

```
::tkutils::tkutodo::widget {path args}
::tkutils::tkutodo::treeWidget {path}
::tkutils::tkutodo::setTodos {path comps}
::tkutils::tkutodo::loadText {path ics}
::tkutils::tkutodo::todos {path}
::tkutils::tkutodo::count {path}
::tkutils::tkutodo::toggle {path {index ""}}
```

21.66 tkutoolbar

Source: lib/tm/tkutils/tkutoolbar-0.2.tm.

```
::tkutils::tkutoolbar::widget {path args}
::tkutils::tkutoolbar::addButton {path id label command args}
::tkutils::tkutoolbar::addToggle {path id label varName args}
::tkutils::tkutoolbar::addDropdown {path id label args}
::tkutils::tkutoolbar::addSeparator {path}
::tkutils::tkutoolbar::addWidget {path id w {side left}}
::tkutils::tkutoolbar::setEnabled {path id enabled}
::tkutils::tkutoolbar::buttonWidget {path id}
::tkutils::tkutoolbar::items {path}
::tkutils::tkutoolbar::configureButton {path id args}
::tkutils::tkutoolbar::setCallback {path id command}
::tkutils::tkutoolbar::setDisplayMode {path mode}
::tkutils::tkutoolbar::getDisplayMode {path}
::tkutils::tkutoolbar::addAction {path name {id ""}}
```

21.67 tkutree

Source: `lib/tm/tkutils/tkutree-0.2.tm`.

```
::tkutils::tkutree::widget {path args}
::tkutils::tkutree::insert {path parent text args}
::tkutils::tkutree::loadTree {path nodes {parent ""}}
::tkutils::tkutree::clear {path}
::tkutils::tkutree::selection {path}
::tkutils::tkutree::itemText {path id}
::tkutils::tkutree::itemValues {path id}
::tkutils::tkutree::children {path {id ""}}
::tkutils::tkutree::treeview {path}
```

21.68 tkuunits

Source: `lib/tm/tkutils/tkuunits-0.1.tm`.

```
::tkutils::tkuunits::units {}
::tkutils::tkuunits::toPixels {value unit {w .}}
::tkutils::tkuunits::fromPixels {px {w .}}
::tkutils::tkuunits::convert {value from to {w .}}
```

21.69 tkuvalidate

Source: `lib/tm/tkutils/tkuvalidate-0.1.tm`.

```
::tkutils::tkuvalidate::attach {w validator args}
::tkutils::tkuvalidate::check {w}
::tkutils::tkuvalidate::valid {w}
::tkutils::tkuvalidate::clear {w}
::tkutils::tkuvalidate::allValid {widgets}
```

21.70 tkuvcard

Source: `lib/tm/tkutils/tkuvcard-0.2.tm`.

```
::tkutils::tkuvcard::widget {path args}
::tkutils::tkuvcard::treeWidget {path}
::tkutils::tkuvcard::loadText {path vcf}
::tkutils::tkuvcard::loadFile {path file}
::tkutils::tkuvcard::setCards {path cards}
::tkutils::tkuvcard::cards {path}
::tkutils::tkuvcard::count {path}
::tkutils::tkuvcard::addCard {path {fn "New Contact"}}
::tkutils::tkuvcard::removeCard {path index}
::tkutils::tkuvcard::addProperty {path cardIndex name value {type ""}}
::tkutils::tkuvcard::removeProperty {path cardIndex propIndex}
::tkutils::tkuvcard::setProperty {path cardIndex propIndex name value
    {type ""}}
::tkutils::tkuvcard::toText {path}
::tkutils::tkuvcard::save {path file}
```

21.71 tkuwheel

Source: lib/tm/tkutils/tkuwheel-0.2.tm.

```
::tkutils::tkuwheel::redirect {target w args}
::tkutils::tkuwheel::unbind {w args}
```

21.72 tkuwinico

Source: lib/tm/tkutils/tkuwinico-0.1.tm.

```
::tkutils::tkuwinico::defaultSizes {}
::tkutils::tkuwinico::fromSvg {svgFile outFile args}
::tkutils::tkuwinico::fromPhoto {photo outFile args}
::tkutils::tkuwinico::fromPhotos {photos outFile}
```

21.73 tkuxml

Source: lib/tm/tkutils/tkuxml-0.1.tm.

```
::tkutils::tkuxml::widget {path args}
::tkutils::tkuxml::setXml {path xml}
::tkutils::tkuxml::loadFile {path filename}
::tkutils::tkuxml::getRoot {path}
```

21.74 tkuzip

Source: lib/tm/tkutils/tkuzip-0.1.tm.

```
::tkutils::tkuzip::widget {path args}
::tkutils::tkuzip::openFile {path zipfile}
::tkutils::tkuzip::getEntries {path}
::tkutils::tkuzip::selectedMember {path}
```

22 Practice: Tk Integration, Measures and Window State

22.1 Load Tk First

`tkutils` complements Tk; it does not replace the Tk runtime. Graphical examples need a display. On Linux a test run under Xvfb can be useful, while a plain `tclsh` without a display creates no windows.

```
package require Tk
package require tkutils::tkuunits

label .label -text "Maßeinheiten"
pack .label -padx 12 -pady 12
puts [::tkutils::tkuunits::toPixels 10 mm .]
```

`tkuunits-0.1.tm` exports `units`, `toPixels`, `fromPixels` and `convert`. The conversion can use the window argument `w`, because the physical screen resolution matters. A pixel value is therefore not the same on all displays.

22.2 Comparing Units

```
package require Tk
package require tkutils::tkuunits

set pixels [::tkutils::tkuunits::toPixels 25.4 mm .]
set inches [::tkutils::tkuunits::convert 25.4 mm inch .]
puts "pixels=$pixels inches=$inches"
```

Do not write the return values as fixed pixel constants into documentation or tests. A good test compares the consistency of two conversions under the same display configuration.

22.3 Window Geometry

`tkugeometry-0.1.tm` exports `save`, `restore`, `track` and `key`. The first three commands expect, among other things, a window and an application name. The implementation also contains logic for adapting to the available screen area.

```
package require Tk
package require tkutils::tkugeometry

set appId demo-window
wm title . "Fensterzustand"
# "." ergibt den Schlüssel main, ohne zu speichern.
set geometryKey [::tkutils::tkugeometry::key .]
puts $geometryKey
```

The example does not write anything persistent. Before using `save`, `restore` and `track`, check storage location, options and behavior with several monitors in particular.

22.4 Troubleshooting Optional Widgets

A widget module may need additional packages, such as Tablelist, Img, tDOM or scrollutil. Test such modules one by one instead of concluding from a successful `package require tkutils` alone that all additional functions are available.

```
package require Tk
set code [catch {package require tkutils::tkutablelist} result options]
if {$code} {
    puts stderr "Tablelist integration unavailable: $result"
} else {
    puts "tkutablelist $result"
}
```

A missing optional package is not an error of the Tk runtime. The application should then either print a clear diagnosis or disable the affected function.

Tested with Tcl/Tk 8.6.14 and 9.0.4 on Linux under Xvfb: all examples of this appendix. With `tk scaling` 1.39 the first example prints `39.38461538461539`, the second `pixels=100.03692307692307 inches=1.0`, the third `main`.

Not tested: different window managers.

23 Test Environment

23.1 Introduction

This appendix names the environment used to check the statements of the book, the behavior of older versions and the points that are not yet checked.

23.2 Environment

What	Version
System	Linux, GUI under Xvfb without a window manager, screen 1280×1024
Tcl/Tk 8.6	8.6.14 with tDOM 0.9.3, sqlite3 3.45.1, tklib 0.8 (Tablelist 6.20, scrollutil 1.17), Img 1.4.16; without tksvg
Tcl/Tk 9	9.0.4 with tDOM 0.9.7, sqlite3 3.53.0, Tablelist 7.11, scrollutil 2.10, Img 2.1.1
tkutils	0.48.0
tclutils	0.68.0
ctrlutils	0.4
tical	1.1, through TICAL_DIR or a sibling folder

The results of the test suite are in Installation, section *Test Run*. For the run, tclutils, tkutils and tical lie side by side as sibling folders; `TCLUTILS_TM`, `TKUTILS_TM` and `TICAL_DIR` point to these trees.

Check scripts: every example with an output `;# -> …` runs under `xvfb-run -a tclsh8.6` and `xvfb-run -a tclsh9.0`, with the modules from `tcl::tm::path add`

`<tclutils>/lib/tm <tkutils>/lib/tm`.

`tkulauncher-9.4` (mouse wheel in scroll mode) was unstable up to `tkulauncher` 0.2: the buttons got the wheel binding only through a later event, and after `reload` not at all. Since 0.3 the launcher binds the buttons when it creates them; `tkulauncher-9.5` checks the case after `reload`.

23.3 Older Versions

23.3.1 Test Suite 0.46.0

Tcl/Tk 8.6.14, Linux, Xvfb, tclutils 0.65.0, tkutils 0.46.0 (commit `401ebde`), tklib 0.8, tDOM 0.9.3, sqlite3 3.45.1, without tksvg: 78 test files, 837 tests, 811 passed, 25 skipped, 1 failed.

- The failure was `tkulauncher-9.4` (mouse wheel in scroll mode). It could not be reproduced afterwards, not even in the unchanged tree (837 tests, 0 failures under 8.6.14 and 9.0.4); `tkulauncher.test` alone passed with 99 of 99 tests. The cause is fixed in `tkulauncher` 0.3 (section *Environment*).
- Skipped were, among others, all 13 tests of `tkumonthcanvas.test` and 7 SVG tests of `tkuwinico.test`.
- 0.47.0 added `tkufiledialog-1.4a` (double click), `tkutltools-2.1`/`-2.2` (both load orders with `tablelist_tile`) and `tkuform-3.7` (option error); the first three fail with the code of 0.46.0.

Without a display. In 0.46.0, `stack.test`, `tkumdview.test`, `tkupdfinspect.test` and `tkuwinico.test` abort on loading without `DISPLAY`, write no summary, and the run ends with exit code 1. Remedy: always start the suite under Xvfb or with a display.

tical. `tkumonthcanvas.test` from 0.46.0 looks only for `tical*/tical` without `TICAL_DIR` and skips all 13 tests in a flat checkout. Remedy: set `TICAL_DIR=/pfad/zu/tical`; then 13 of 13 pass.

23.3.2 Two Tablelist Variants up to 0.46.0

`tkutablelist` loads `tablelist_tile` first and `Tablelist` only when that is missing. The `tkutl*` helpers, in contrast, load `tablelist`. With Tablelist 6.20 from tklib 0.8, `package require tkutils::tkutltools` (and `tkutlsort` alone) therefore fails after `package require tkutils` with

```
can't set "usingTile": it is not supported to use both Tablelist and
    Tablelist_tile in the same application
```

Remedy: `package require tablelist` (or `tkutils::tkutltools`) **before** the umbrella. Then `tkutablelist` takes the non-ttk variant, and both families run together.

23.3.3 tkufiledialog 0.1 (tkutils 0.46.0)

Dialog operated programmatically in each case (Tcl/Tk 8.6.14, Linux, Xvfb):

Operation	Result
type the name a.txt, Return	full path to a.txt
Escape	""

Operation	Result
save -initialfile bericht -defaultextension .txt, OK	…/bericht.txt
directory, OK without selection	the start folder
double-click handler on file a.txt	""

The last case is a bug in the code of 0.46.0: the double click on a file enters the name and closes with *OK*, but does not set the return value. Remedy: select the file and use *OK* or Return. Fixed in 0.47.0 (`tkufiledialog` 0.2).

23.4 Not Yet Checked

Chapter	open
all	Windows and macOS
Dialogs	appearance, focus and placement on a real screen
File Widgets	navigation of a file manager with Windows paths
Geometry and Units	configuration folder on Windows and macOS; several monitors; zoom state (Xvfb without a window manager has none)
Actions and Layout	starting app entries on Windows (only read in the code)
Designer and load	the app Dateiblick (code not in tclutils, tkutils or ctrlutils)
tkdevtools	display of the encoding profiles under Tcl 9; font aliases on Windows

A result for one of these points belongs in the respective chapter and is removed here.

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